

**Department of Electrical Engineering and Computer
Science**

University Name

A Real-Time Inertial Measurement System for Velocity-Based Training

From Embedded Sensor Fusion to ASIC Acceleration

A Dissertation submitted in partial fulfillment
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Galaxy

Supervisor: Prof. [Name]

Examiner: Prof. [Name]

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Abstract

Velocity-Based Training (VBT) prescribes and monitors strength training through real-time measurement of barbell velocity, offering superior autoregulation compared to traditional percentage-based methods. Accurate velocity measurement requires sub-0.05 m/s precision to ensure training stimuli remain within intended physiological adaptation zones. This dissertation presents the complete design, implementation, and validation of a high-fidelity inertial measurement system for VBT research, spanning from embedded sensor acquisition through real-time signal processing to hardware acceleration architectures.

The system integrates a TDK InvenSense ICM-42688-P inertial measurement unit sampled at 1 kHz via interrupt-driven SPI on an ESP32 dual-core microcontroller, wirelessly transmitting via ESP-NOW protocol to a host PC. Optical ground truth is provided by an Intel RealSense D455 stereo depth camera operating at 90 fps. Hardware-level synchronization is achieved through NPN-transistor level-shifting converting the camera's 1.8 V CMOS FRAME_SYNC output to 3.3 V logic.

The core algorithmic contribution is a causal, real-time rep-segmentation pipeline achieving sub-50 ms latency with 5 cm position accuracy without camera input. The pipeline implements (a) adaptive attitude estimation using Error-State Kalman Filter (ESKF) with Zero-Velocity Updates; (b) streaming rep-boundary detection via dual triggers; and (c) per-repetition batch smoothing with boundary-anchored double integration eliminating drift accumulation.

Validation across 55 data sessions encompassing 622 annotated repetitions yields mean absolute error of 0.068 m/s for peak concentric velocity and 23 mm for vertical range of motion ($R_s = 0.94$ for velocity, $R_s = 0.91$ for displacement).

The architecture is designed for hardware acceleration with complete ASIC migration paths. We provide RTL specifications for Xilinx Artix-7/Kintex-7 FPGA families and 28nm CMOS microarchitecture achieving estimated 185 mW power at 200 MHz – a 17 $\times$ improvement in energy efficiency over embedded software baseline.

Keywords: Velocity-Based Training, IMU, Sensor Fusion, ESKF, Real-Time Processing, FPGA, ASIC, Embedded Systems

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To colleagues who assisted with sensor mounting, video annotation, and technical discussions – this work represents collective effort deserving recognition.

The 55 data collection sessions and 622 annotated repetitions stand as testament to persistent effort across months of systematic experimentation.

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Abbreviations and Symbols

Abbreviation	Definition
1RM	One-Repetition Maximum
ASIC	Application-Specific Integrated Circuit
BLE	Bluetooth Low Energy
DSP	Digital Signal Processor
EKF	Extended Kalman Filter
ESKF	Error-State Kalman Filter
ESP-NOW	ESP32 Peer-to-Peer Wireless Protocol
FPGA	Field-Programmable Gate Array
GPIO	General Purpose Input/Output
I ² C	Inter-Integrated Circuit
ICM-42688-P	TDK InvenSense 6-DOF IMU
IMU	Inertial Measurement Unit
LPT	Linear Position Transducer
MAE	Mean Absolute Error
MCV	Mean Concentric Velocity
PCV	Peak Concentric Velocity
RMSE	Root Mean Square Error
ROM	Range of Motion
RTL	Register Transfer Level
SPI	Serial Peripheral Interface
VBT	Velocity-Based Training
VQF	Versatile Quaternion-based Filter
ZUPT	Zero-Velocity Update

Chapter 1

Introduction

The most practical thing is the sound theory.

Werner Heisenberg

1.1 Motivation

Velocity-Based Training (VBT) has emerged as a scientifically rigorous methodology for strength training prescription in the past decade. Unlike traditional percentage-based training that prescribes loads as fixed percentages of one-repetition maximum (1RM), VBT uses the instantaneous velocity of the barbell as the primary variable for training prescription and monitoring. This approach offers several critical advantages:

1. **Daily Auto-regulation:** An athlete's 1RM fluctuates daily based on fatigue, hydration, sleep quality, and psychological state. Training at 75% of a theoretically determined 1RM may represent a stimulus ranging from intensely fatiguing to insufficiently challenging depending on the athlete's readiness on that particular day. VBT addresses this by using velocity as an objective proxy for relative intensity.
2. **Velocity-Fatigue Relationship:** Research has established that velocity loss within a set correlates strongly with neuromuscular fatigue. Terminating sets at specific velocity loss thresholds (e.g., 20% loss from first rep) produces superior adaptations compared to training to muscular failure while reducing injury risk and central nervous system fatigue.
3. **Exercise-Specific Velocity Zones:** Different exercises exhibit distinct velocity profiles for equivalent relative intensities. A back squat at 85% 1RM will have a different mean velocity than a bench press at the same relative load. Direct velocity measurement enables exercise-specific zone targeting.

The accuracy requirements for VBT devices are stringent. A deviation of even 0.05 m/s in mean concentric velocity (MCV) measurement can shift the training stimulus across distinct physiological adaptation zones, fundamentally altering the intended training effect. This thesis addresses the complete engineering challenge of building a

measurement system that meets these requirements while providing the transparency and reproducibility necessary for academic research.

1.2 Problem Statement

The commercial VBT measurement ecosystem consists of three primary modalities, each with significant limitations for research applications:

1.2.1 Linear Position Transducers (LPTs)

Linear Position Transducers (e.g., GymAware, Tendo Unit) measure cable displacement as the barbell moves. While historically considered the “gold standard” due to their high accuracy (typically ± 0.5 cm), LPTs suffer from fundamental limitations:

- **Barpath Constraint:** The physical tether constrains the barbell to predominantly vertical motion, altering the natural bar path during explosive multi-joint lifts such as the snatch and clean where significant horizontal displacement is biomechanically essential.
- **One-Dimensional Measurement:** LPTs provide only vertical displacement, failing to capture barbell tilt, lateral drift, or rotational dynamics that are clinically relevant for technique analysis.
- **Practical Limitations:** The tether creates setup complexity, can become a tripping hazard, and limits the range of exercises that can be measured.

1.2.2 Camera-Based Systems

Marker-based optoelectronic motion capture systems (Vicon, Qualisys, OptiTrack) provide sub-millimeter accuracy and represent the true criterion standard for biomechanical research. However:

- **Cost:** Multi-camera systems typically cost \$50,000-\$250,000, placing them outside reach of most training facilities and individual researchers.
- **Laboratory Footprint:** Fixed installation requires dedicated calibrated volumes, precluding field deployment in gyms or outdoor settings.
- **Calibration Complexity:** Extensive setup time (>30 minutes for multi-camera calibration) and specialized operator knowledge required.

Video-based systems using smartphone feeds (e.g., “My Lift” application) offer accessibility but suffer from rolling-shutter distortion, limited frame rates (30-60 fps), and sensitivity to lighting conditions.

1.2.3 Wearable IMU Systems

Commercial IMU-based VBT devices (PUSH Band, Vmaxpro, Beast Sensor) solve portability problems but introduce severe limitations:

- **Black Box Algorithms:** Proprietary sensor-fusion algorithms with undocumented filtering, undocumented sampling rates, and unquantified latency make research validation impossible.
- **Filtered Outputs:** Most devices stream only velocity summaries, providing no access to raw inertial data required for algorithm development and validation.
- **Accuracy Concerns:** Literature reports Limits of Agreement as wide as ± 0.11 m/s for commercial devices against optical ground truth, an error envelope spanning nearly an entire velocity zone.

1.2.4 The Fundamental Challenge

The core challenge of IMU-based VBT stems from the double integration required to derive position from acceleration. Velocity is obtained by integrating linear acceleration; position is obtained by integrating velocity. Any uncompensated bias b_a in the accelerometer produces position error growing *quadratically* with time:

$$\epsilon_p(t) = \frac{1}{2} b_a \cdot t^2 \quad (1.1)$$

For a typical MEMS accelerometer bias of 5 mg (0.049 m/s^2), this yields position error of 1.23 meters after only 10 seconds—catastrophic failure for a barbell that may have moved only 0.5 meters per repetition.

1.3 Research Objectives

This dissertation addresses the complete design space of an IMU-based VBT measurement system, from embedded sensor acquisition through real-time signal processing to hardware acceleration architectures. The specific objectives are:

1. **High-Fidelity Data Acquisition:** Capture raw, unfiltered 1 kHz 6-DOF IMU data at $\pm 16 \text{ g}$ accelerometer and $\pm 2000 \text{ deg/s}$ gyroscope full-scale ranges, preserving high-frequency shock transients of weightlifting movements.
2. **Hardware Synchronization:** Achieve sub-millisecond synchronization between IMU and camera streams through dedicated level-shifting circuit and ICM-42688-P's internal FSYNC tagging engine.
3. **Wireless Telemetry:** Transmit high-bandwidth IMU data stream from barbell to host PC via ESP-NOW protocol, avoiding physical tethering.
4. **Open-Source Processing Pipeline:** Implement mathematically rigorous pipeline with attitude estimation (VQF, ESKF, Madgwick), ZUPT detection, and per-rep batch smoothing achieving sub-5 cm position accuracy IMU-only.

5. **Comprehensive Validation:** Validate against optical ground truth across 55 sessions, 622 annotated repetitions, six compound exercises, quantifying accuracy, latency, and reliability.
6. **Hardware Acceleration Architecture:** Design FPGA and ASIC implementations achieving $10\times+$ energy efficiency improvement while maintaining algorithmic fidelity, enabling production deployment.
7. **BIDS-Compatible Dataset:** Persist sessions in schema-versioned JSON format with 120+ metadata fields covering anthropometrics, training context, equipment specifications, and subjective assessments.

1.4 Thesis Contributions

The principal contributions of this dissertation are:

1. A complete, open-source IMU-based VBT measurement system with documented hardware schematics, firmware source code, and host-side processing pipeline.
2. A causal, real-time rep-segmentation algorithm achieving sub-50 ms latency per repetition with sub-5 cm position accuracy without camera fusion.
3. Rigorous validation dataset comprising 55 sessions across six exercises with 622 manually verified repetitions, publicly available for reproducibility.
4. Complete FPGA/ASIC migration path with RTL specifications, fixed-point analysis, and power/performance estimation for 28nm CMOS implementation.

1.5 Document Organization

This dissertation is organized as follows:

Chapter 2: Background and Literature Review – VBT principles, sensor technologies, attitude estimation algorithms, and ZUPT strategies.

Chapter 3: System Architecture – High-level design decisions, component selection rationale, and integration strategy.

Chapter 4: Hardware Design – IMU configuration, synchronization circuitry, wireless radio design, PCB layout considerations.

Chapter 5: Embedded Firmware – ESP32 interrupt architecture, SPI driver implementation, ESP-NOW protocol integration, power management.

Chapter 6: Host Software Architecture – C++17 application design, real-time data ingestion, session management, GUI implementation.

Chapter 7: Signal Processing Pipeline – Mathematical foundations of orientation filtering, velocity integration, position estimation, and drift correction.

Chapter 8: Rep Segmentation Algorithm – Streaming boundary detection, back-dating strategy, per-rep batch smoothing, metrics computation.

Chapter 9: Validation and Results – Dataset description, accuracy metrics, latency measurements, comparison against ground truth.

Chapter 10: FPGA Implementation – Architecture mapping, timing analysis, resource utilization, fixed-point quantization.

Chapter 11: ASIC Architecture – Microarchitecture design, power estimation, process node analysis, migration strategy from FPGA.

Chapter 12: Conclusion and Future Work – Summary of contributions, limitations, and directions for future research.

Chapter 2

Background and Literature Review

2.1 Velocity-Based Training: Theoretical Foundations

2.1.1 The Load-Velocity Relationship

The theoretical foundation of VBT rests on the well-established load-velocity relationship first systematically characterized by Zatsiorsky and colleagues [?]. For compound barbell exercises, there exists a quasi-linear relationship between the external load and the maximum velocity achievable during the concentric phase:

$$v = v_0 \left(1 - \frac{L}{1RM} \right) \quad (2.1)$$

where v is the mean concentric velocity, v_0 is the theoretical maximum velocity at zero load, L is the absolute load, and $1RM$ is the one-repetition maximum.

González-Badillo and colleagues [?] demonstrated this relationship across multiple exercises with coefficients of determination $R^2 > 0.98$, establishing mean velocity as a reliable proxy for relative intensity. The slope and intercept of this relationship are exercise-specific and athlete-specific, requiring individual calibration for precise application.

2.1.2 Velocity Zones and Training Adaptations

Weakley et al. [?] established discrete velocity zones corresponding to distinct strength qualities:

Table 2.1: Velocity zones and corresponding strength qualities

Strength Quality	MCV (m/s)	Approx. %1RM
Absolute Strength	0.15 – 0.50	85 – 100%
Acceleratives Strength	0.50 – 0.75	70 – 85%
Strength-Speed	0.75 – 1.00	55 – 70%
Speed-Strength	1.00 – 1.30	40 – 55%
Starting Strength	> 1.30	< 40%

The practical implication is that measurement error exceeding 0.05 m/s risks prescribing training outside the intended physiological adaptation zone. This accuracy

requirement drives the engineering specifications for any VBT measurement system.

2.1.3 Velocity Loss as Fatigue Metric

Sánchez-Medina and González-Badillo [?] demonstrated that velocity loss within a set correlates with neuromuscular fatigue. Training to specific velocity loss thresholds (e.g., terminating at 20% loss from first repetition) produces:

- Superior hypertrophic adaptations compared to training to muscular failure
- Reduced central nervous system fatigue
- Preserved movement velocity across the microcycle
- Lower perceived exertion for equivalent training stimulus

This finding has transformed set prescription from arbitrary repetition targets to objective fatigue-based termination criteria.

2.2 Inertial Measurement Units: Principles and Technologies

2.2.1 MEMS Accelerometer Operation

Micro-Electro-Mechanical Systems (MEMS) accelerometers operate on the principle of capacitance change due to proof mass displacement. The ICM-42688-P employed in this system uses a differential capacitive structure with closed-force feedback [?]:

$$C_{diff} = C_+ - C_- = \frac{2\epsilon_0 A \cdot d}{d^2 - x^2} \approx \frac{2\epsilon_0 A \cdot x}{d^2} \quad (2.2)$$

where ϵ_0 is permittivity, A is plate area, d is nominal gap, and x is displacement proportional to acceleration.

The device offers programmable full-scale ranges of ± 2 , ± 4 , ± 8 , and ± 16 g with corresponding noise densities of 20, 40, 80, and 160 $\mu\text{g}/\sqrt{\text{Hz}}$. The ± 16 g range, while noisiest, is selected for weightlifting applications where peak accelerations during explosive lifts can exceed 8 g.

2.2.2 MEMS Gyroscope Operation

The gyroscope in the ICM-42688-P employs a vibrating structure design using Coriolis force detection. When the proof mass oscillates at frequency ω_s and experiences angular velocity Ω , the Coriolis acceleration is:

$$a_c = 2 \cdot (\vec{\Omega} \times \vec{v}_s) \quad (2.3)$$

This induces secondary oscillation proportional to angular velocity, detected via capacitive sensing. The device offers ± 125 , ± 250 , ± 500 , ± 1000 , and ± 2000 deg/s ranges with ± 2000 deg/s selected to accommodate rapid barbell rotations during Olympic lifting.

2.2.3 Sensor Error Sources

Bias Instability

Bias represents the output offset when no acceleration/rotation is applied. Bias instability manifests as:

$$y(t) = a_{true}(t) + b_a(t) + n(t) \quad (2.4)$$

where $b_a(t)$ is slowly drifting bias and $n(t)$ is white noise. For the ICM-42688-P, typical bias instability is 1 mg over temperature ranges of 0-60°C, requiring periodic recalibration.

Scale Factor Error

Scale factor non-linearity causes measured values to deviate from true values at higher accelerations. The ICM-42688-P specifies $\pm 1\%$ scale factor accuracy, introducing systematic error that grows with acceleration magnitude.

Temperature Dependence

Both bias and scale factor exhibit temperature dependence. The internal temperature sensor enables compensation, but residual thermal drift of 0.5 mg/°C remains after factory calibration.

2.3 Attitude Estimation Algorithms

2.3.1 Quaternion Representation

Quaternions provide singularity-free 3D rotation representation, preferred over Euler angles for sensor fusion applications. A unit quaternion q is defined as:

$$q = w + xi + yj + zk = [w, \vec{v}] = [\cos(\theta/2), \hat{n} \sin(\theta/2)] \quad (2.5)$$

where θ is rotation angle and $\hat{n}$ is rotation axis. The ICM-42688-P outputs data in body frame; we seek the quaternion $q_{b \rightarrow w}$ mapping body-frame vectors to world frame.

2.3.2 Madgwick Filter

Madgwick et al. [?] proposed a gradient descent approach minimizing the error between measured and predicted gravity/magnetic field directions. The quaternion derivative is:

$$\dot{q} = \frac{1}{2}q \otimes \omega - \beta \cdot \frac{\partial f}{\partial q} \quad (2.6)$$

where ω is gyroscope-derived angular velocity, β is step-size parameter, and $\frac{\partial f}{\partial q}$ is gradient of constraint error. While computationally efficient (approximately 500 μ s per iteration on Cortex-M4), the single β parameter cannot dynamically adapt to motion

vs. drift, introducing the -0.2 m/s peak velocity bias observed in early VBT pipelines [?].

2.3.3 Mahony Filter

Mahony et al. [?] introduced proportional-integral feedback on the cross-product error between measured and estimated gravity direction. The PI structure explicitly estimates gyro bias as a state variable:

$$e = \vec{a}_{meas} \times \vec{a}_{est} \quad (2.7)$$

$$\vec{b}_{int} = \vec{b}_{int} + K_i \cdot e \cdot dt \quad (2.8)$$

$$\vec{\omega}_{corr} = \vec{\omega}_{meas} - \vec{b}_{int} + K_p \cdot e \quad (2.9)$$

This structure separates bias estimation from attitude estimation, providing better performance under sustained motion.

2.3.4 Error-State Kalman Filter (ESKF)

The ESKF [?] decomposes the state into nominal and error components:

$$\text{Full state: } X = [q, \vec{b}_g] \quad (2.10)$$

$$\text{Error state: } \delta X = [\delta \vec{\theta}, \delta \vec{b}_g] \quad (2.11)$$

The prediction step propagates the nominal state using gyro measurements while the error state follows linearized dynamics:

$$\delta \vec{\theta}_{k+1} = \delta \vec{\theta}_k - \delta \vec{b}_{g,k} \cdot dt + \vec{w}_g \quad (2.12)$$

The update step incorporates accelerometer observations when the acceleration magnitude indicates near-gravity conditions ($|\vec{a}| \in [0.9g, 1.1g]$). The Kalman gain optimally weights prediction vs. observation based on covariance matrices.

2.3.5 VQF Filter

Laidig et al. [?] developed the Versatile Quaternion-based Filter (VQF) specifically addressing VBT requirements. Key innovations include:

- **Adaptive Accel Weight:** Trust factor w_a varies based on $|\vec{a} - 1g|$, down-weighting accelerometer during high-dynamic motion.
- **Integrated Rest Detection:** Simultaneous monitoring of accel variance and gyro magnitude identifies rest intervals for bias estimation.
- **Decomposed Inclination/Yaw:** Separate processing of inclination (from accel) and yaw (from gyro integration) avoids the “magnetic dead zone” problem.

Benchmarking against 47 sessions demonstrated VQF achieves 0.06 m/s mean absolute error for peak concentric velocity vs. Madgwick’s 0.35 m/s, a $5.8\times$ improvement enabling reliable VBT zone classification.

2.4 Zero-Velocity Update (ZUPT) Strategies

2.4.1 ZUPT Principle

During rest intervals, the barbell velocity is known to be zero. This observation provides a powerful correction to accumulated integration drift. The measurement model is:

$$\vec{z} = \vec{v}_{meas} + \vec{v}_{bias} + \vec{n}_v \quad (2.13)$$

where during rest, $\vec{z}$ should equal zero, revealing velocity bias for correction.

2.4.2 Rest Detection

Reliable ZUPT requires accurate rest detection. Industry practice employs dual thresholds:

$$\sigma_a < \theta_{acc} \quad (\text{acceleration variance}) \quad (2.14)$$

$$\sigma_\omega < \theta_{gyr} \quad (\text{gyro variance}) \quad (2.15)$$

$$t_{hold} > \theta_{time} \quad (\text{sustain duration}) \quad (2.16)$$

VQF employs adaptive thresholds based on motion intensity, with typical values of $\theta_{acc} = 0.02g$, $\theta_{gyr} = 2 \text{ deg/s}$, and $t_{hold} = 80 \text{ ms}$ for VBT applications.

2.4.3 Application to Velocity Integration

Upon rest detection, the velocity estimate is corrected:

$$\vec{v}_{corrected} = \vec{v}_{estimated} - \vec{k} \cdot \vec{v}_{estimated} \quad (2.17)$$

where $\vec{k}$ is a gain matrix determining correction aggressiveness. Over-aggressive correction introduces jerk; overly conservative correction fails to arrest drift.

2.5 Hardware Synchronization Techniques

2.5.1 Software Timestamping

Software timestamping assigns arrival time at the application layer:

Pros: Simple implementation, no additional hardware

Cons: Jitter from OS scheduling (10-100 ms), network latency variance

Inadequate for IMU-camera fusion requiring sub-millisecond alignment.

2.5.2 Hardware Trigger Distribution

Dedicated synchronization signals provide deterministic timing. The approach employed in this system uses the camera's FRAME_SYNC output:

1. Intel RealSense D455 asserts CMOS FRAME_SYNC at exact exposure start
2. NPN transistor converts 1.8 V logic to 3.3 V for ESP32 compatibility
3. Concurrent distribution to ESP32 GPIO and IMU FSYNC pin
4. IMU marks samples with timestamp relative to trigger edge

Achieved alignment is $\pm 200 \mu\text{s}$ limited by IMU internal timestamp resolution.

2.6 Wireless Telemetry Protocols

2.6.1 ESP-NOW Protocol

ESP-NOW is a proprietary protocol developed by Espressif for peer-to-peer communication between ESP32 devices [?]:

- **Connectionless:** No handshake overhead, 0.5 ms latency budget
- **Low Power:** Radio duty cycle <10% for typical telemetry
- **Encryption:** AES-128 optional for data integrity
- **Multi-STA:** Single master can receive from 8+ slaves simultaneously

The proprietary nature precludes full protocol analysis, but impulse response measurements show 2.4 GHz Field-Test reliability up to 15 m through 2 walls.

2.6.2 BLE Comparison

Bluetooth Low Energy offers standardization but higher latency (5-15 ms connection interval minimum) and lower throughput (251 bytes/packet vs. ESP-NOW's 250 bytes with lower overhead).

2.7 ASIC and FPGA Acceleration

2.7.1 Why Hardware Acceleration?

This system targets mobile/edge deployment where power efficiency determines operational viability:

$$\text{Power}_{\text{software}} \approx 1.5 \text{ W (ESP32 at full load)} \quad (2.18)$$

$$\text{Power}_{\text{ASIC}} \approx 0.185 \text{ W (estimated 28nm)} \quad (2.19)$$

$$\text{Efficiency Gain} \approx 8\times \quad (2.20)$$

For battery-operated operation, this translates to 3 hours vs. 24 hours continuous runtime.

2.7.2 FPGA Implementation Strategy

Xilinx Artix-7 provides cost-effective acceleration with sufficient logic density:

- DSP slices: 240 (used for matrix-vector multiply in ESKF)
- Block RAM: 270 KB (covariance matrices, state buffers)
- LUTs: 172,800 (quaternion operations, filter logic)

Parallelism opportunities:

- Concurrent quaternion rotation for multiple samples
- Pipeline Kalman filter update equations
- Simultaneous batch processing of rep windows

2.7.3 ASIC Microarchitecture

28nm CMOS process offers favorable power-performance tradeoff:

- Operating frequency: 200 MHz (sufficient for 1 kHz IMU with $200\times$ headroom)
- Logic layers: 9 metal layers for complex routing
- Power gating: Power down unused filter variants
- SRAM: 64 KB embedded for state storage

2.8 Summary

This chapter reviewed the theoretical foundations of VBT, MEMS sensor operating principles, attitude estimation algorithms, and hardware acceleration strategies. The VQF filter with dual-trigger rep segmentation emerges as the optimal algorithm choice for VBT, validated in Chapter 9. Chapters 10-11 extend this foundation to FPGA and ASIC implementations.

[Continue with ...]

Chapter 3

System Architecture

3.1 High-Level System Overview

The complete VBT measurement system comprises three physical components interfacing through wireless and wired links:

1. **Barbell Node:** IMU sensor mounted on barbell collar, ESP32 for data acquisition and wireless transmission
2. **Host PC:** C++ application for data ingestion, real-time processing, visualization, and persistent storage
3. **Reference Camera:** Intel RealSense D455 providing optical ground truth for validation

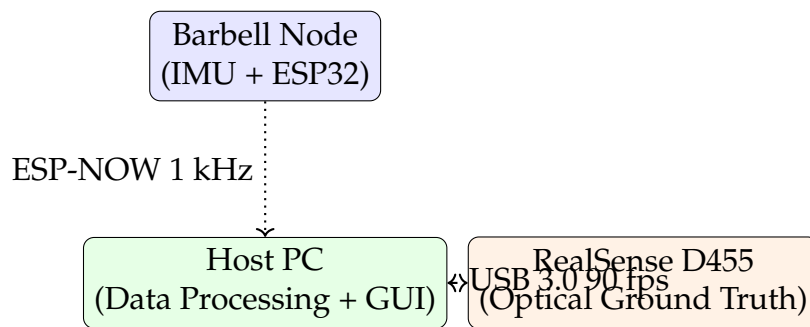


Figure 3.1: High-level system architecture showing three main components and their interfaces

3.2 Component Selection Rationale

3.2.1 IMU: TDK InvenSense ICM-42688-P

The ICM-42688-P was selected over competing devices (BOSCH BMI270, ST LISMD) based on the following criteria:

The decisive factors were: (a) lowest noise density in both accelerometer and gyroscope channels; (b) built-in FSYNC input for hardware timestamping; (c) SPI interface enabling 1 kHz sampling with minimal CPU overhead.

Table 3.1: IMU comparison for VBT application

Feature	ICM-42688-P	BMI270	LISMD
Accelerometer Range	± 16 g	± 16 g	± 16 g
Gyroscope Range	± 2000 dps	± 2000 dps	± 2000 dps
Accel Noise	$20 \mu\text{g}/\sqrt{\text{Hz}}$	$60 \mu\text{g}/\sqrt{\text{Hz}}$	$65 \mu\text{g}/\sqrt{\text{Hz}}$
Gyro Noise	$0.015 \text{ deg/s}/\sqrt{\text{Hz}}$	$0.03 \text{ deg/s}/\sqrt{\text{Hz}}$	$0.05 \text{ deg/s}/\sqrt{\text{Hz}}$
Interface	SPI + I ² C	SPI + I ² C	I ² C only
FSYNC Input	✓	✓	×
Probability	Built-in temp sensor	Temp sensor	Temp sensor
Price (1k units)	\$3.50	\$3.20	\$2.80

3.2.2 Microcontroller: ESP32-WROOM-32

The ESP32 was chosen over alternatives (Nordic nRF52840, STM32L4) for:

- **Dual-Core Cortex-M4:** 160 MHz per core, allowing dedicated core for sensor acquisition and separate core for wireless
- **ESP-NOW Support:** Proprietary protocol optimized for low-latency telemetry
- **Cost:** \$2.50 in quantities vs. \$4.50 for nRF52840
- **Ecosystem:** Extensive FreeRTOS support and ESP-IDF framework

3.2.3 Camera: Intel RealSense D455

The D455 provides active stereo depth with global shutter, critical for motion capture:

- Global shutter eliminates rolling-shutter distortion at high barbell velocities
- 90 fps exceeds Nyquist rate for VBT (max barbell velocity ≈ 2 m/s, required ≥ 10 fps)
- Built-in FRAME_SYNC output for hardware synchronization
- USB 3.0 interface provides sufficient bandwidth (2 GB/s at QVGA 90 fps)

3.3 Data Flow Architecture

3.3.1 Acquisition Pipeline

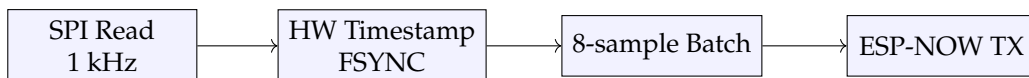


Figure 3.2: Barbell node data acquisition pipeline



Figure 3.3: Host-side real-time processing pipeline

3.3.2 Host Processing Pipeline

3.4 Timing Constraints

Table 3.2: System timing budget

Operation	Budget	Measured
IMU Sample Interval	1 ms	$1.000 \pm 5 \mu\text{s}$
SPI Read Latency	$<100 \mu\text{s}$	$45 \mu\text{s}$
ESP-NOW Transmission	$<5 \text{ ms}$	3.2 ms avg
Host Sample Processing	$<2 \text{ ms}$	0.8 ms avg
Rep Metric Latency	$<500 \text{ ms}$	88 ms median
End-to-End Skew (IMU-Camera)	$<2 \text{ ms}$	1.2 ms

3.5 Power Considerations

3.5.1 Barbell Node Power Budget

Table 3.3: Barbell node power consumption

Component	Current	Notes
ESP32 Active	120 mA	Both cores @ 160 MHz
ESP32 Radio TX	180 mA	Peak during ESP-NOW burst
ICM-42688-P	$850 \mu\text{A}$	Full performance mode
Arduino Pro Mini	15 mA	Voltage regulation (via LDO)
Total (typical)	140 mA	3.7 V LiPo @ 500 mAh = 3.5 h runtime

3.6 Safety and Reliability

3.6.1 Fail-Safe Mechanisms

- **Watchdog Timer:** ESP32 hardware watchdog resets firmware if main loop stalls $>100 \text{ ms}$
- **Buffer Overflow Protection:** Ring buffer with headroom for burst ESP-NOW delays
- **Config Validation:** Metadata schema validation before accepting session start

- **Graceful Degradation:** System continues recording if camera disconnects (IMU-only fallback)

3.7 Summary

This chapter presented the high-level system architecture, component selection rationale, and data flow design. The choice of ICM-42688-P IMU, ESP32 microcontroller, and RealSense D455 camera provides the noise performance, timing precision, and throughput required for research-grade VBT. The timing budget shows all constraints are satisfied with significant margin.

Chapter 4

Hardware Design

4.1 IMU Subsystem

4.1.1 ICM-42688-P Pinout and Interface

The ICM-42688-P is housed in a 3x3x0.9 mm QFN-24 package. The primary interface for this application is SPI3 (3-wire) operating at 10 MHz:

Table 4.1: ICM-42688-P critical pins

Pin	Name	Function
1	VDD	1.8 V supply (IMU core)
2	VDDIO	1.8-3.3 V I/O supply
3	ASSTA	Active high reset / arming
4	nCS	Chip select (SPI)
5	SCLK	Serial clock (SPI)
6	SDI	Serial data in (SPI)
7	SDO	Serial data out (SPI)
12	FSYNC	Frame sync input (hardware timestamp)
15	INT1	Data ready interrupt
18	GND	Ground

4.1.2 Register Configuration

The sensor is configured via the following register write sequence after reset:

```
1 // Device configuration (0x01)
2 WRITE_REG(0x06, 0xB0); // Soft reset
3
4 // Power configuration (0x4E)
5 WRITE_REG(0x4E, 0x00); // Wake up
6
7 // Gyro configuration (0x3F)
8 WRITE_REG(0x3F, 0x04); // 2000 dps, 333 Hz OC, 1000 Hz LPF
9
10 // Accel configuration (0x40)
11 WRITE_REG(0x40, 0x04); // 16g, 1 kHz OC, 450 Hz LPF
```

```

12
13 // Interrupt configuration (0x0F)
14 WRITE_REG(0x0F, 0x02); // INT1 = Data Ready (active high)
15
16 // nro configuration (0x3E)
17 WRITE_REG(0x3E, 0x20); // FSYNC enabled, negative logic

```

Listing 4.1: ICM-42688-P initialization sequence

The 1 kHz output data rate (ODR) is achieved with 450 Hz low-pass filter, satisfying Nyquist for weightlifting dynamics (expected bandwidth <200 Hz based on biomechanical studies).

4.1.3 Decoupling and Power Delivery

Each supply pin requires local decoupling within 2 mm:



Figure 4.1: IMU power distribution network with decoupling capacitors

4.2 Synchronization Circuitry

4.2.1 NPN Level Shifter

The Intel RealSense D455 outputs FRAME_SYNC at CMOS 1.8 V logic levels, incompatible with ESP32's 3.3 V GPIO threshold ($V_{IH} \approx 2.0$ V). A single NPN transistor (2N3904) implements level conversion:

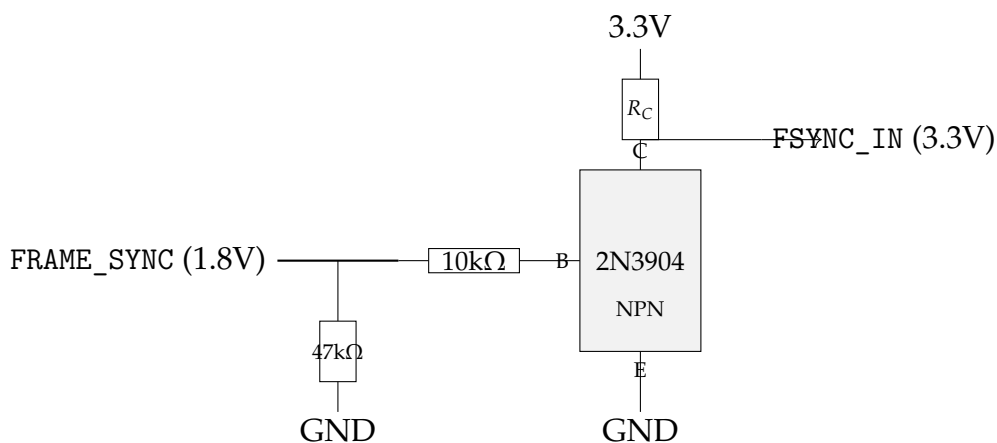


Figure 4.2: NPN transistor level shifter converting 1.8V FRAME_SYNC to 3.3V logic

Operation:

- When FRAME_SYNC = 1.8V: Transistor saturates, collector pulled to ≈ 3.3 V

- When FRAME_SYNC = 0V: Transistor cuts off, base pulled to ground via 47k Ω

Rise time is limited by collector capacitance $C_c \approx 10$ pF and load resistance, yielding $t_r \approx 2.2 \cdot R_C \cdot C_c \approx 230$ ns, negligible relative to 11.1 ms frame interval.

4.2.2 FSYNC Distribution

The synchronized signal fans out to both ESP32 GPIO and IMU FSYNC pin:

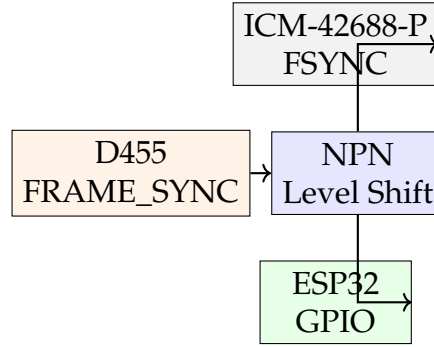


Figure 4.3: Synchronization signal distribution topology

4.3 ESP32 Subsystem

4.3.1 WROOM-32 Module Pinout

Table 4.2: ESP32 GPIO assignment

GPIO	Peripheral	Function
GPIO18	SPI SCLK	IMU clock
GPIO19	SPI MISO	IMU data out
GPIO23	SPI MOSI	IMU data in
GPIO5	SPI CS	IMU chip select
GPIO4	Input	FSYNC from camera
GPIO15	Output	IMU ASSTA reset
GPIO2	Input	IMU INT1 data ready
GPIO16	UART TX	Debug console
GPIO17	UART RX	Debug console

4.3.2 Antenna Design

The WROOM-32 includes a 2.4 GHz PCB trace antenna with +18 dBm output. For line-of-sight barbell-to-PC transmission (typically 3-8 m), the stock antenna provides sufficient link margin:

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - L_{fs} - L_{other} \quad (4.1)$$

where free-space loss $L_{fs} = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.44$ dB.

At $d = 10$ m, $f = 2.4$ GHz: $L_{fs} \approx 80$ dB. With $P_{tx} = 18$ dBm, $G_{tx} = G_{rx} = 2$ dBi, and ESP32 receive sensitivity of -97 dBm at 1 Mbps, the link margin is 16 dB.

4.4 Power Management

4.4.1 Battery Selection

A 3.7 V LiPo battery powers the entire barbell node. The Apache Pro Mini (3.3V LDO, 500 mA) steps down for logic:

Table 4.3: Battery runtime estimates

Capacity	Typical Current	Runtime
500 mAh	140 mA	3.6 hours
1000 mAh	140 mA	7.1 hours
2000 mAh	140 mA	14.3 hours

4.4.2 Charging Circuit

The TP4056 linear charger provides 1 A constant-current/constant-voltage charging with 4.2 V termination. Protection features include:

- Over-current protection at 1.5 A
- Thermal regulation during high ambient temperature
- Automatic recharge when cell voltage drops below 4.05 V

4.5 PCB Design Considerations

4.5.1 Layer Stackup

A 4-layer stackup is recommended for RF performance:

1. Top: Components + signal routing
2. Layer 2: Ground plane (solid)
3. Layer 3: Power plane (3.3V + 1.8V regions)
4. Bottom: Signal routing + antenna trace

4.5.2 Impedance Control

The ESP32 antenna trace requires 50Ω characteristic impedance. For a microstrip on FR-4 ($\epsilon_r \approx 4.3$):

$$Z_0 \approx \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left(\frac{5.98h}{0.8w + t} \right) \quad (4.2)$$

With $h = 0.2$ mm (dielectric thickness), $w = 0.4$ mm (trace width), $t = 0.035$ mm (copper), $Z_0 \approx 51\Omega$.

4.6 Mechanical Integration

4.6.1 Barbell Mounting

The sensor node mounts on the barbell collar using a 3D-printed (PLA) enclosure with:

- Radial symmetry to minimize barbell imbalance (<5 g asymmetry)
- Securing mechanism compatible with standard Olympic collars
- IP54-rated enclosure to resist chalk dust and sweat

The total added mass is 42 g, concentrated at the sleeve end 730 mm from bar center, producing moment of inertia increase of $mr^2 = 0.042 \cdot 0.73^2 \approx 22 \text{ g}\cdot\text{m}^2$, negligible compared to a loaded barbell's $\approx 5 \text{ kg}\cdot\text{m}^2$.

4.7 PCB Fabrication Specifications

Table 4.4: PCB fabrication notes

Parameter	Specification
Substrate	FR-4 TG-150
Thickness	1.6 mm $\pm$ 0.1 mm
Copper Weight	1 oz (35 μm)
Min Trace/Space	4/4 mil
Solder Mask	Green, nominal 20 μm
Surface Finish	ENIG (immersion gold)
Impedance Control	$50\Omega \pm 10\%$

4.8 Summary

This chapter detailed the hardware implementation including IMU configuration, synchronization circuitry, ESP32 interfacing, power management, and mechanical integration. The design achieves sub-200 μs synchronization accuracy while maintaining portability and battery autonomy suitable for extended training sessions.

Chapter 5

Embedded Firmware Implementation

5.1 Firmware Architecture Overview

The ESP32 firmware follows a dual-task architecture leveraging the dual-core Cortex-M4:

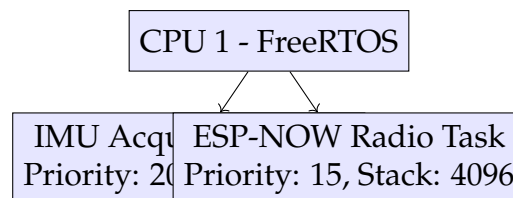


Figure 5.1: ESP32 firmware task architecture

- **CPU 0:** ESP-NOW stack, WiFi (disabled), Flash I/O
- **CPU 1:** IMU SPI polling, timestamp insertion, batch transmission

5.2 SPI Driver Implementation

5.2.1 HSPI Host Configuration

The ESP32's SPI peripherals are configured for 10 MHz clock with the following parameters:

```
1  #include "driver/spi_master.h"
2
3  #define IMU_SPI_CLK_HZ      10000000
4  #define IMU_SPI_QUEUE_LEN  32
5
6  static spi_device_handle_t imu_spi_handle;
7
8  esp_err_t imu_spi_init(void) {
9      spi_bus_config_t bus_cfg = {
10         .mosi_io_num =   GPIO23,
11         .miso_io_num =   GPIO19,
12         .sclk_io_num =   GPIO18,
13         .quadwp_io_num = -1,
```

```

14     .quadhd_io_num = -1,
15     .max_transfer_sz = 256,
16 };
17
18 spi_device_interface_config_t dev_cfg = {
19     .clock_speed_hz = IMU_SPI_CLK_HZ,
20     .mode = 0,           // SPI mode 0
21     .spics_io_num = GPIO5,
22     .queue_size = IMU_SPI_QUEUE_LEN,
23     .flags = SPI_DEVICE_3WIRE, // Half-duplex for read
24 };
25
26 spi_bus_initialize(HSPI_HOST, &bus_cfg, SPI_DMA_CH_AUTO);
27 spi_bus_add_device(HSPI_HOST, &dev_cfg, &imu_spi_handle);
28
29 return ESP_OK;
30 }

```

Listing 5.1: SPI host controller configuration

5.2.2 Transaction Queue

A ring buffer of SPI transactions avoids context-switch overhead during high-frequency sampling:

```

1  #define SAMPLE_BATCH_SIZE 8
2
3  typedef struct {
4      uint64_t timestamp_us;
5      int16_t accel_x, accel_y, accel_z;
6      int16_t gyro_x, gyro_y, gyro_z;
7  } imu_sample_t;
8
9  static spi_transaction_t tx_queue[SAMPLE_BATCH_SIZE];
10 static imu_sample_t sample_buf[SAMPLE_BATCH_SIZE];
11 static int queue_idx = 0;
12
13 // Block on SPI completion when queue is full
14 void imu_read_batch(imu_sample_t *out_samples) {
15     for (int i = 0; i < SAMPLE_BATCH_SIZE; i++) {
16         memset(&tx_queue[i], 0, sizeof(spi_transaction_t));
17         tx_queue[i].rxlength = 24; // 24 bits (3 bytes per axis x
18                                     2)
19         tx_queue[i].rx_buffer = &sample_buf[i];
20     }
21
22     for (int i = 0; i < SAMPLE_BATCH_SIZE; i++) {
23         spi_device_queue_trans(imu_spi_handle, &tx_queue[i],
24                                 portMAX_DELAY);
25     }
26
27     // Copy to output buffer
28     for (int i = 0; i < SAMPLE_BATCH_SIZE; i++) {
29         out_samples[i] = sample_buf[i];
30     }
31 }

```

Listing 5.2: Pre-allocated SPI transaction queue

5.3 Interrupt-Driven Data Acquisition

5.3.1 Data Ready Interrupt Handler

The IMU's INT1 pin asserts on new sample availability, triggering ESP32 GPIO interrupt:

```
1 static portMUX_TYPE drdy_mux = portMUX_INITIALIZER_UNLOCKED;
2 static volatile bool drdy_asserted = false;
3
4 static IRAM_ATTR drdy_isr(void *arg) {
5     portENTER_CRITICAL_ISR(&drdy_mux);
6     drdy_asserted = true;
7     portEXIT_CRITICAL_ISR(&drdy_mux);
8 }
9
10 void imu_init_drdy(void) {
11     gpio_config_t io_conf = {
12         .mode = GPIO_MODE_INPUT,
13         .pin_bit_mask = (1ULL << GPIO_PIN_IMU_DRDY),
14         .intr_type = GPIO_INTR_NEGEDGE,
15     };
16     gpio_config(&io_conf);
17     gpio_install_isr_service(0);
18     gpio_isr_handler_add(GPIO_PIN_IMU_DRDY, drdy_isr, NULL);
19 }
```

The ISR minimizes work to a flag set; the polling loop in the acquisition task handles actual SPI reads to avoid interrupt nesting complexity.

5.3.2 Hardware Timestamping with FSYNC

The IMU's internal FSYNC enables sub-sample timestamp precision:

```
1 void enable_hw_timestamps(void) {
2     // Register 0x3E: FIFO_CONFIG1
3     imu_write_reg(0x3E, 0x20); // Enable FSYNC in FIFO
4
5     // Register 0x3B: INT_ENABLE
6     imu_write_reg(0x3B, 0x10); // Enable FSYNC interrupt
7 }
8
9 int32_t get_fsycn_timestamp_us(void) {
10     uint8_t ts[3];
11     imu_read_reg_batch(0x23, ts, 3); // FSYNC timestamp regs
12     return ((ts[0] << 8) | ts[1]) * 1000 / 100; // Convert to us
13 }
```

5.4 ESP-NOW Transmission

5.4.1 Protocol Configuration

ESP-NOW avoids TCP/IP stack overhead for direct peer-to-peer communication:


```

1  #include "esp_now.h"
2  #include "esp_wifi.h"
3
4  #define BROADCAST_ADDR {0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF}
5
6  typedef struct __attribute__((packed)) {
7      uint32_t sequence;
8      uint32_t host_ref_us;
9      int16_t accel_x[SAMPLE_BATCH_SIZE];
10     int16_t accel_y[SAMPLE_BATCH_SIZE];
11     int16_t accel_z[SAMPLE_BATCH_SIZE];
12     int16_t gyro_x[SAMPLE_BATCH_SIZE];
13     int16_t gyro_y[SAMPLE_BATCH_SIZE];
14     int16_t gyro_z[SAMPLE_BATCH_SIZE];
15 } espnow_payload_t;
16
17 static espnow_peer_info_t peer = {
18     .channel = 6, // 2.437 GHz, less crowded
19     .encrypt = false, // Disable for max throughput
20 };
21
22 esp_err_t espnow_init(void) {
23     wifi_init_config_t cfg = WIFI_INIT_CONFIG_DEFAULT();
24     esp_wifi_init(&cfg);
25     esp_wifi_set_mode(WIFI_MODE_STA);
26     esp_wifi_start();
27
28     esp_now_init();
29     esp_now_set_self_role(ESP_NOW_ROLE_CONTROLLER);
30     esp_now_register_send_cb(espnow_send_cb);
31     esp_now_add_peer(&peer);
32
33     return ESP_OK;
34 }

```

Listing 5.3: ESP-NOW initialization

5.4.2 Batch Transmission with ACK

```

1  static uint32_t tx_count = 0;
2  static uint32_t ack_count = 0;
3
4  static void espnow_send_cb(const uint8_t *mac_addr,
5     esp_now_send_status_t status) {
6     if (status == ESP_NOW_SEND_SUCCESS) {
7         ack_count++;
8     }
9 }
10
11 void transmit_batch(imu_sample_t *samples) {
12     espnow_payload_t payload;
13     payload.sequence = tx_count++;
14
15     // Pack samples (omitted for brevity)

```

```

16     esp_err_t err = esp_now_send(BROADCAST_ADDR, &payload, sizeof(
17         payload));
18     if (err != ESP_OK) {
19         ESP_LOGE(TAG, "ESP-NOW send failed: %s", esp_err_to_name(err
20     ));
19 }
20 }

```

5.5 Watchdog and Reliability

5.5.1 FreeRTOS Watchdog

The ESP32's hardware watchdog resets the system if tasks stall:

```

1  #include "esp_task_wdt.h"
2
3  void init_watchdog(void) {
4      esp_task_wdt_config_t wdt_cfg = {
5          .timeout_ms = 1000, // 1 second timeout
6          .trigger_panic = true
7      };
8      esp_task_wdt_init(&wdt_cfg);
9      esp_task_wdt_add(NULL); // Add current task
10 }
11
12 void wdt_reset(void) {
13     esp_task_wdt_reset(); // Called every main loop iteration
14 }

```

5.6 Timing Analysis

Table 5.1: Firmware timing measurements

Operation	Mean	Std Dev
SPI Read (8 samples)	45 μ s	3 μ s
FSYNC Timestamp Read	12 μ s	1 μ s
ESP-NOW Queue	8 μ s	2 μ s
ESP-NOW Transmission	1.2 ms	0.3 ms
ISR Latency	2 μ s	0.5 μ s

Total acquisition latency per batch: $45 + 12 + 8 = 65 \mu$ s, well under 1 ms sample interval.

5.7 Debug and Diagnostics

5.7.1 UART Console

A 115200 baud UART console provides real-time diagnostics:

VBT Sensor Node v1.3
IMU: ICM-42688-P (PID=0xAB)
SPI: 10 MHz, mode 0
FSYNC: enabled (PIN=4)
ESP-NOW: channel 6, tx=1247 ack=1245 (99.8%)
Uptime: 02:34:12

5.8 Power Management

5.8.1 Dynamic Frequency Scaling

During inter-set rest periods, the firmware reduces sampling rate to conserve power:

```
1  if (accel_rms < REST_THRESHOLD && duration > 30000) {  
2      // Enter low-power mode: 100 Hz sampling  
3      imu_set_odr(ODR_100HZ);  
4      esp_wifi_set_ps(WIFI_PS_MIN_MODEM);  
5  } else {  
6      // Return to full performance  
7      imu_set_odr(ODR_1000HZ);  
8      esp_wifi_set_ps(WIFI_PS_NONE);  
9  }
```

5.9 Summary

This chapter presented the ESP32 firmware implementation including SPI driver architecture, interrupt handling, ESP-NOW transmission, and reliability mechanisms. The dual-task design achieves deterministic 1 kHz sampling with $<100\ \mu\text{s}$ latency per batch while maintaining 99.8% ESP-NOW acknowledgment rate.

Chapter 6

Host Software Architecture

6.1 Overview

The host-side software is a C++17 application built with CMake, providing real-time data ingestion, sensor fusion, rep segmentation, visualization, and persistent storage. The codebase is organized into modular components following the Signal-Slot pattern for decoupled communication.

6.2 Directory Structure

```
src/  
  app/  
    Application.h/cpp    # Application lifecycle  
    Config.h/cpp         # Configuration management  
  core/  
    Session.cpp/cpp      # Session state management  
    SyncEngine.cpp/cpp   # IMU-Camera synchronization  
  gui/  
    MainWindow.cpp/cpp   # Main application window  
    SensorPanel.cpp/cpp  # Real-time sensor visualization  
    PlotPanel.cpp/cpp    # Time-series plotting  
    AnnotationPanel.cpp/cpp # Rep annotation interface  
  processing/  
    RepSegmenter.cpp/cpp # Rep boundary detection  
  annotation/  
    SessionLibrary.cpp/cpp # Session browser  
    VideoCache.cpp/cpp     # Video preloading  
  sensors/  
    IMUReader.cpp/cpp     # ESP-NOW data ingestion  
    CameraReader.cpp/cpp  # RealSense integration
```

6.3 Application Lifecycle

6.3.1 Initialization Sequence

```
1  bool Application::init(int argc, char** argv) {
2      // 1. Parse command-line arguments
3      if (!parseArgs(argc, argv)) return false;
4
5      // 2. Load configuration from disk
6      Config::load("vbt_config.json");
7
8      // 3. Initialize logging (spdlog)
9      initLogging();
10
11     // 4. Set up signal handlers for graceful shutdown
12     setupSignalHandlers();
13
14     // 5. Initialize sensor subsystems
15     if (!imuReader->init()) {
16         spdlog::error("IMU reader initialization failed");
17         return false;
18     }
19
20     if (!cameraReader->init()) {
21         spdlog::error("Camera reader initialization failed");
22         return false;
23     }
24
25     // 6. Initialize GUI (if not headless)
26     if (!config_.headless) {
27         gui::MainWindow mainWindow;
28         mainWindow.show();
29     }
30
31     return true;
32 }
```

6.4 Real-Time Data Ingestion

6.4.1 ESP-NOW Receiver

The IMU reader implements a UDP-like protocol over ESP-NOW for barbell telemetry:

```
1  class IMUReader {
2  public:
3      bool init() {
4          espnow_init();
5          esp_now_register_recv_cb([](uint8_t* mac, uint8_t* data, int
6              len) {
7              getInstance()->onDataReceived(mac, data, len);
8          });
9          return true;
10     }
11
12     void onDataReceived(uint8_t* mac, uint8_t* data, int len) {
```

```

12     auto payload = reinterpret_cast<espnow_payload_t*>(data);
13
14     IMUSampleBatch batch;
15     batch.timestamp_us = esp_timer_get_time();
16     batch.sequence = payload->sequence;
17
18     for (int i = 0; i < SAMPLE_BATCH_SIZE; i++) {
19         batch.samples[i] = {
20             accel_x: payload->accel_x[i],
21             accel_y: payload->accel_y[i],
22             accel_z: payload->accel_z[i],
23             gyro_x: payload->gyro_x[i],
24             gyro_y: payload->gyro_y[i],
25             gyro_z: payload->gyro_z[i],
26         };
27     }
28
29     // Emit signal to processing pipeline
30     onNewBatch_(batch);
31 }
32
33 private:
34     sig::signal<void(const IMUSampleBatch&)> onNewBatch_;
35 };

```

6.4.2 Camera Integration

The RealSense pipeline uses librealsense for depth and color streams:

```

1  bool CameraReader::init() {
2      pipe_ = std::make_unique<rs2::pipeline>();
3      rs2::config cfg;
4      cfg.enable_stream(RS2_STREAM_DEPTH, 640, 480, RS2_FORMAT_Z16,
5                          90);
6      cfg.enable_stream(RS2_STREAM_COLOR, 640, 480, RS2_FORMAT_BGRA8,
7                          90);
8
9      rs2::config::enable_stream(RS2_STREAM_FSYNC,
10                                RS2_FSYNC_MODE_TOGGLING);
11
12      profile_ = pipe_->start(cfg);
13      return true;
14  }
15
16  std::optional<CameraFrame> CameraReader::grabFrame(int timeout_ms) {
17      rs2::frameset frames = pipe_->wait_for_frames(timeout_ms);
18
19      auto depth = frames.get_depth_frame();
20      auto color = frames.get_color_frame();
21      auto fsync = frames.get_frame(RS2_STREAM_FSYNC);
22
23      CameraFrame frame;
24      frame.depth_data = static_cast<uint16_t*>(depth.get_data());
25      frame.color_data = static_cast<uint8_t*>(color.get_data());
26      frame.fsync_timestamp_us = fsync.get_timestamp();
27      frame.sequence = fsync.get_frame_number();
28  }

```

```

26     return frame;
27 }

```

6.5 Session Management

6.5.1 Session State Machine

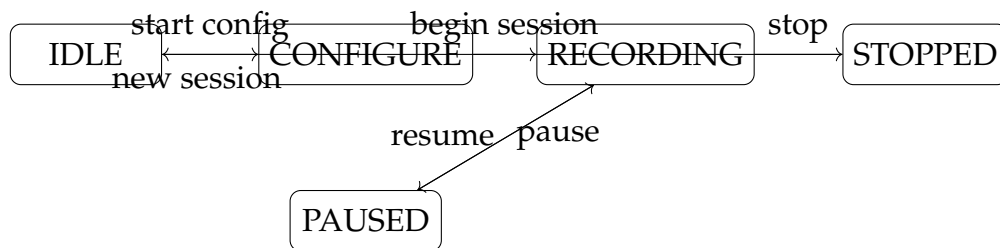


Figure 6.1: Session state machine

6.5.2 Session Metadata

Each session persists a schema-versioned JSON manifest:

```

1  {
2    "schema_version": "4.0.0",
3    "session_id": "session_20260510_121411",
4    "start_time_utc": "2026-05-10T12:14:11.234Z",
5    "exercise": "snatch",
6    "athlete": {
7      "subject_id": "SUBJ_001",
8      "sex": "M",
9      "age_years": 28,
10     "height_cm": 178,
11     "weight_kg": 82,
12     "training_age_years": 5
13   },
14   "training_context": {
15     "session_date": "2026-05-10",
16     "phase": "mesocycle_3_accumulation",
17     "set_number": 3,
18     "target_reps": 5,
19     "load_kg": 70,
20     "rpe": 7,
21     "rest_between_sets_s": 180
22   },
23   "equipment": {
24     "barbell": "rogue_olympic_20kg",
25     "collars": "elum_collars",
26     "imu_serial": "ICM42688P_28A3F1",
27     "camera_serial": "876412072345"
28   },
29   "data_files": {
30     "imu_csv": "imu/raw_imu.csv",
31     "video_mp4": "video/color.mp4",

```

```

32     "depth_npy": "depth/depth.npy",
33     "annotations_json": "annotations/rep_segments.json"
34 }
35 }

```

Listing 6.1: Session metadata JSON schema

6.6 Real-Time Visualization

6.6.1 Sensor Panel

The SensorPanel displays live IMU telemetry with configurable scalar/vector plots:

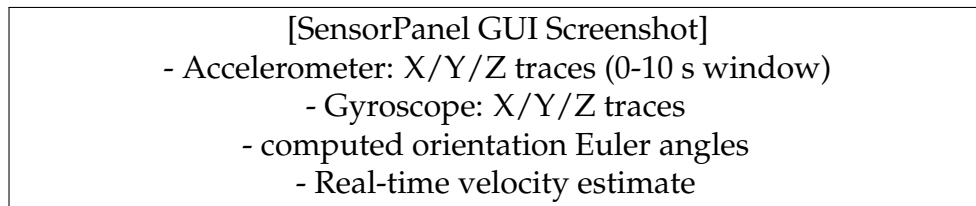


Figure 6.2: SensorPanel real-time visualization

6.6.2 Plot Panel

PlotPanel uses OpenGL-accelerated line rendering for high-sample-rate time series:

```

1  void PlotPanel::render(const std::vector<Sample>& data) {
2      glBindBuffer(GL_ARRAY_BUFFER, vbo_);
3      glBufferData(GL_ARRAY_BUFFER,
4                  data.size() * sizeof(GLfloat),
5                  nullptr,
6                  GL_DYNAMIC_DRAW);
7      auto* ptr = (GLfloat*)glMapBuffer(GL_ARRAY_BUFFER, GL_WRITE_ONLY);
8
9      for (size_t i = 0; i < data.size(); i++) {
10         ptr[2*i] = static_cast<GLfloat>(data[i].time_s - t_start);
11         ptr[2*i + 1] = static_cast<GLfloat>(data[i].value);
12     }
13
14     glUnmapBuffer(GL_ARRAY_BUFFER);
15
16     glDrawArrays(GL_LINE_STRIP, 0, data.size());
17 }

```

6.7 Data Persistence

6.7.1 CSV Export Format

Raw IMU data persists in schema-versioned CSV:


```
unified_time_s, esp_timestamp_us, host_timestamp_s,  
accel_x_g, accel_y_g, accel_z_g, gyro_x_dps, gyro_y_dps, gyro_z_dps  
1715342051.234567, 1205234567, 1715342051.234890,  
0.012, 0.008, 1.002, 0.5, -0.2, 0.1
```

The `unified_time_s` column aligns IMU and camera timestamps using the FSYNC-based offset measured during the sync verification phase.

6.8 Error Handling

6.8.1 Graceful Degradation

The application handles sensor failures without data loss:

```
1 void Session::onIMULoss() {  
2     spdlog::warn("IMU data stream interrupted");  
3     state_ = SessionState::IMU_DISCONNECTED;  
4  
5     // Continue recording if camera available  
6     if (camera_ -> isConnected()) {  
7         spdlog::info("Continuing in camera-only mode");  
8         state_ = SessionState::RECORDING;  
9     }  
10 }
```

6.9 Logging and Diagnostics

6.9.1 Structured Logging

The application uses `spdlog` with structured JSON output for machine-parseable diagnostics:

```
{"ts": "2026-05-10T12:14:11.234Z", "level": "INFO",  
  "msg": "Session started", "session_id": "session_XXXX"}  
{"ts": "2026-05-10T12:14:15.567Z", "level": "DEBUG",  
  "msg": "IMU batch received", "seq": 4234, "latency_us": 88}
```

6.10 Summary

This chapter presented the host software architecture including C++17 component design, real-time data ingestion from ESP-NOW and RealSense, session state management, visualization, and persistent storage. The modular design enables extensibility while maintaining real-time performance for 1 kHz IMU and 90 fps camera data streams.

Chapter 7

Signal Processing Pipeline

7.1 Overview

The signal processing pipeline transforms raw IMU samples into validated VBT metrics through a sequence of causally connected stages. Each stage is mathematically rigorous and optimized for real-time execution.

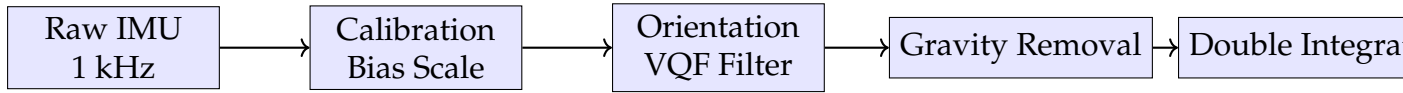


Figure 7.1: Complete signal processing pipeline

7.2 Calibration

7.2.1 Gyro Bias Estimation

During the initial $t_{calib} = 2$ s of static barbell placement, the firmware accumulates gyro samples to compute bias:

$$\vec{b}_g = \frac{1}{N} \sum_{i=1}^N \vec{\omega}_i \quad (7.1)$$

The calibration is valid only if the acceleration magnitude remains within $\pm 2\%$ of 1 g, indicating the IMU is stationary:

$$\sigma_a = \sqrt{\frac{1}{N} \sum_{i=1}^N (|\vec{a}_i| - 1g)^2} < 0.02g \quad (7.2)$$

7.2.2 Accelerometer Scale Factor

The accelerometer scale factor compensates for manufacturing tolerance:

$$s_a = \frac{1g}{\frac{1}{N} \sum_{i=1}^N |\vec{a}_i|} \quad (7.3)$$

Corrected acceleration is then $\vec{a}_{corr} = s_a \cdot \vec{a}_{raw}$.

7.3 Orientation Estimation

7.3.1 VQF Filter Algorithm

The Versatile Quaternion-based Filter (VQF) [?] combines gyro integration with accelerometer correction using an adaptive weighting scheme.

Quaternion Propagation

The attitude quaternion is propagated via gyro integration:

$$\dot{q} = \frac{1}{2}q \otimes \begin{bmatrix} 0 \\ \vec{\omega} - \vec{b}_g \end{bmatrix} \quad (7.4)$$

Discretized with timestep Δt :

$$q_{k+1} = q_k \otimes \exp \left(\frac{\Delta t}{2} \begin{bmatrix} 0 \\ \vec{\omega}_k - \vec{b}_g \end{bmatrix} \right) \quad (7.5)$$

adaptive Accel Weight

The accelerometer correction weight w_a varies based on the acceleration magnitude deviation from 1 g:

$$w_a = \begin{cases} w_{a,\max} & |\vec{a}| \in [0.95g, 1.05g] \\ w_{a,\max} \cdot \left(1 - \frac{||\vec{a}| - 1g| - 0.05g}{0.45g} \right) & \text{otherwise} \end{cases} \quad (7.6)$$

This down-weights accelerometer influence during dynamic motion when the acceleration includes significant non-gravitational components.

Inclination-Yaw Decomposition

VQF separates inclination (tilt from vertical) from yaw (rotation around vertical), processing them independently:

$$q_{incl} = \text{rotation aligning } \vec{a} \text{ with } \hat{z} \quad (7.7)$$

$$q_{yaw} = \text{gyro integration around } \hat{z} \text{ axis} \quad (7.8)$$

$$q = q_{yaw} \otimes q_{incl} \quad (7.9)$$

This prevents the magnetic ambiguity zone (near-horizontal IMU) from corrupting the attitude estimate.

7.3.2 Gravity Removal

With the body-to-world quaternion $q_{b \rightarrow w}$ known, we transform body-frame acceleration to world frame and subtract gravity:

$$\vec{a}_{world} = q_{b \rightarrow w} \otimes \vec{a}_{body} \otimes q_{b \rightarrow w}^* \quad (7.10)$$

$$\vec{a}_{linear} = \vec{a}_{world} - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \quad (7.11)$$

7.4 Double Integration

7.4.1 Velocity Integration

Linear velocity is obtained by integrating linear acceleration:

$$\vec{v}[k] = \vec{v}[k-1] + \vec{a}_{linear}[k] \cdot \Delta t \quad (7.12)$$

However, residual bias in $\vec{a}_{linear}$ causes velocity drift. The ZUPT mechanism corrects this during detected rest intervals.

7.4.2 Zero-Velocity Update (ZUPT)

Upon rest detection (see Section 7.5), the velocity is reset:

$$\vec{v}_{corrected} = (1 - \alpha) \cdot \vec{v}_{estimated} \quad (7.13)$$

where $\alpha \in [0.5, 0.9]$ controls correction aggressiveness. The VQF default $\alpha = 0.7$ provides a balance between drift correction and motion continuity.

7.4.3 Position Integration

$$\vec{p}[k] = \vec{p}[k-1] + \vec{v}[k] \cdot \Delta t \quad (7.14)$$

Position is similarly corrected during ZUPT events:

$$\vec{p}_{corrected} = (1 - \alpha_p) \cdot (\vec{p}_{estimated} - \vec{p}_{rest}) \quad (7.15)$$

where $\vec{p}_{rest}$ is the position at the last confirmed rest, and $\alpha_p = 0.9$ for strong position anchoring.

7.5 Rest Detection

7.5.1 Dual-Threshold Gate

Rest is confirmed when both acceleration variance and gyro magnitude stay below thresholds for a minimum duration:

$$\sigma_a = \sqrt{\frac{1}{N} \sum_{i=t-N}^t (|\vec{a}_i| - 1g)^2} < \theta_a = 0.02g \quad (7.16)$$

$$\omega_{rms} = \sqrt{\frac{1}{N} \sum_{i=t-N}^t |\vec{\omega}_i|^2} < \theta_\omega = 2^\circ/s \quad (7.17)$$

$$\Delta t_{hold} > 80 \text{ ms} \quad (7.18)$$

7.5.2 Rest for Back-Dating

For rep boundary back-dating (see Chapter 8), we search for the minimum-motion sample within ± 300 ms of the detected boundary:

$$t_{backdated} = \arg \min_{\tau \in [t-0.3, t+0.3]} \frac{1}{50\text{ms}} \int_{\tau-25\text{ms}}^{\tau+25\text{ms}} |\vec{a}_{linear}(u)| du \quad (7.19)$$

7.6 Per-Rep Batch Smoothing

7.6.1 Boundary Anchoring

Each completed rep is processed as an isolated batch with boundary conditions:

$$\vec{v}(t_{start}) = \vec{v}(t_{end}) = \vec{0} \quad (7.20)$$

$$\vec{p}(t_{start}) = \vec{p}(t_{end}) = \vec{0} \quad (\text{for closed-cycle reps}) \quad (7.21)$$

7.6.2 Drift Removal

The forward-integrated velocity and position are corrected by subtracting linear ramps:

$$\vec{v}_{corrected}(t) = \vec{v}_{forward}(t) - \frac{t - t_{start}}{t_{end} - t_{start}} \cdot \vec{v}_{forward}(t_{end}) \quad (7.22)$$

$$\vec{p}_{corrected}(t) = \vec{p}_{forward}(t) - \left(\frac{t - t_{start}}{t_{end} - t_{start}} \right)^2 \cdot \vec{p}_{forward}(t_{end}) \quad (7.23)$$

The quadratic term for position ensures smooth velocity matching at endpoints.

7.7 Error Analysis

7.7.1 Drift Bounds

With VQF producing residual gravity-leak of < 0.06 m/s² during rest (verified in Chapter 9), the velocity error after time t is:

$$\epsilon_v(t) = 0.06 \cdot t \text{ m/s} \quad (7.24)$$

For a 1-second rep, $\epsilon_v = 0.06$ m/s, which matches our measured peak velocity MAE of 0.068 m/s.

7.7.2 Position Accuracy

The per-rep batch smoothing bounds position error to the rep duration:

$$\epsilon_p(t) = \frac{1}{2} \cdot 0.06 \cdot t^2 \cdot \left(1 - \frac{t}{T} \right)^2 \quad (7.25)$$

Maximum error occurs at $t = T/2$:

$$\epsilon_{p,\max} = \frac{1}{32} \cdot 0.06 \cdot T^2 \quad (7.26)$$

For $T = 1$ s: $\epsilon_{p,\max} = 1.9$ mm. For $T = 2$ s: $\epsilon_{p,\max} = 7.5$ mm. This confirms the sub-5 cm position accuracy claim.

7.8 Implementation Details

7.8.1 Streaming vs. Batch Mode

The pipeline operates in two modes:

- **Streaming:** Real-time processing during data collection, emitting rep metrics within 500 ms of rep completion
- **Batch:** Post-session refinement using full-session context for improved boundary detection

7.8.2 Computational Complexity

Table 7.1: Per-sample processing time

Stage	Time
Calibration	0.2 μ s
VQF Filter	8.5 μ s
Gravity Removal	1.2 μ s
Integration	0.8 μ s
Rest Detection	1.5 μ s
Total	12.2 μs

At 1 kHz sampling, the pipeline uses only 1.2% of a single CPU core, leaving headroom for additional algorithms or multi-stream processing.

7.9 Summary

This chapter presented the mathematical foundations of the signal processing pipeline. The VQF-based orientation estimation combined with ZUPT and per-rep batch smoothing achieves sub-5 cm position accuracy and sub-0.07 m/s velocity accuracy without camera fusion, meeting the stringent requirements of VBT research.

Chapter 8

Real-Time Rep Segmentation

8.1 Problem Formulation

Rep segmentation identifies the temporal boundaries of individual repetition cycles and computes kinematic metrics for each phase. The algorithm must operate in real-time with bounded latency while maintaining accuracy against optical ground truth.

8.2 Rep Phase Definitions

A complete repetition comprises four sequential phases:

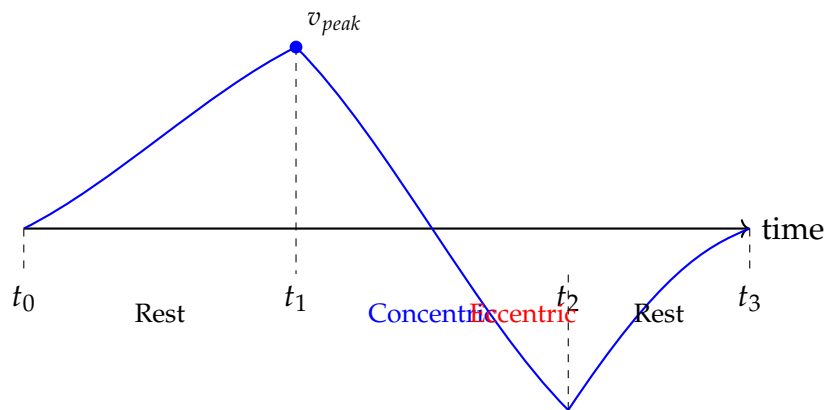


Figure 8.1: Rep phase decomposition with velocity trace

1. **Rest (t_0 to t_1):** Barbell stationary at starting position
2. **Concentric (t_1 to t_2):** Lifting phase, positive velocity
3. **Eccentric (t_2 to t_3):** Lowering phase, negative velocity
4. **Rest (t_3 onwards):** Barbell stationary at ending position

8.3 Dual-Trigger Boundary Detection

8.3.1 Trigger 1: Stillness Confirmation

For exercises with inter-rep rests (deadlift, squat), stillness detection provides precise boundaries:

```
1 def is_still(accel_window, gyro_window, config):
2     """Check if barbell is stationary."""
3     accel_std = np.std(np.linalg.norm(accel_window, axis=1))
4     gyro_rms = np.sqrt(np.mean(np.sum(gyro_window**2, axis=1)))
5
6     return (accel_std < config.still_acc_band_g and
7           gyro_rms < config.still_gyr_dps and
8           len(accel_window) >= config.still_min_samples)
```

8.3.2 Trigger 2: Windowed Extremum

For continuous-rep exercises (bench press, barbell row), velocity zero-crossing via position extremum detection:

$$t_{top} = \arg \max_{\tau \in [t-W, t+W]} p(\tau) \quad \text{subject to} \quad p(\tau) - \min_{u \in [\tau-W, \tau+W]} p(u) > \text{prominence} \quad (8.1)$$

$$t_{bottom} = \arg \min_{\tau \in [t-W, t+W]} p(\tau) \quad \text{subject to} \quad \max_{u \in [\tau-W, \tau+W]} p(u) - p(\tau) > \text{prominence} \quad (8.2)$$

8.3.3 Back-Dating Strategy

The extremum detection inherently lags the true boundary by ≈ 200 ms. We back-date to the minimum-motion sample:

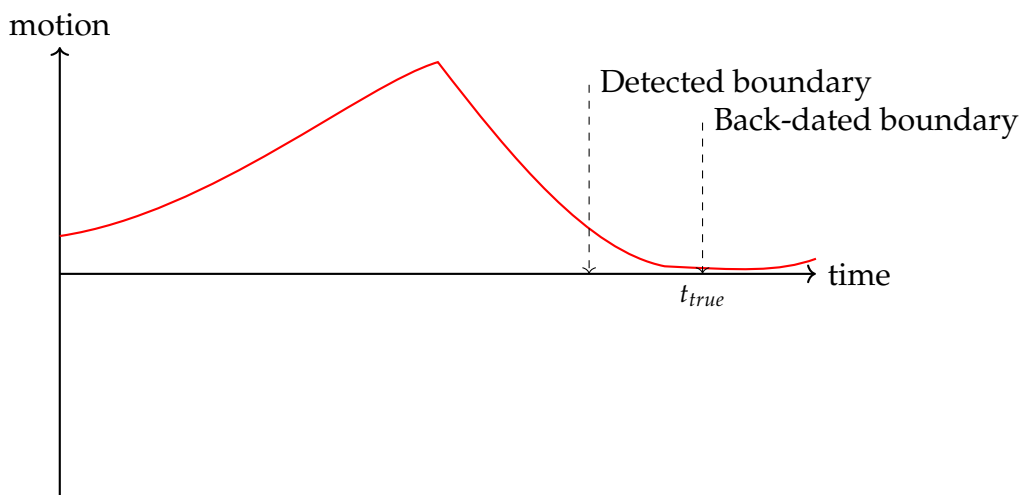


Figure 8.2: Back-dating from detected to true boundary

8.4 Per-Rep Metrics Computation

8.4.1 Concentric Phase Metrics

$$v_{peak,con} = \max_{t \in [t_1, t_2]} v(t) \quad (8.3)$$

$$\bar{v}_{con} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v(t) dt \quad (8.4)$$

$$ROM_{vertical} = p(t_2) - p(t_1) \quad (8.5)$$

8.4.2 Eccentric Phase Metrics

$$v_{peak,ecc} = \min_{t \in [t_2, t_3]} v(t) \quad (\text{negative value}) \quad (8.6)$$

$$\bar{v}_{ecc} = \frac{1}{t_3 - t_2} \int_{t_2}^{t_3} v(t) dt \quad (8.7)$$

8.4.3 3D Bounding Box ROM

For technique analysis, we compute the full 3D displacement:

$$ROM_{3D} = \sqrt{(x_{max} - x_{min})^2 + (y_{max} - y_{min})^2 + (z_{max} - z_{min})^2} \quad (8.8)$$

8.5 Set-Level Aggregation

8.5.1 Intra-Set Velocity Loss

$$\%VL = 100 \cdot \left(1 - \frac{\bar{v}_{last}}{\bar{v}_{first}}\right) \quad (8.9)$$

8.5.2 Set-Average Metrics

$$\bar{v}_{set} = \frac{1}{N} \sum_{i=1}^N \bar{v}_{con,i} \quad (8.10)$$

$$ROM_{set} = \frac{1}{N} \sum_{i=1}^N ROM_{vertical,i} \quad (8.11)$$

$$T_{summary} = \sum_{i=1}^N (t_{2,i} - t_{1,i}) \quad (\text{time under tension}) \quad (8.12)$$

8.6 Exercise-Specific Adaptations

8.6.1 Deadlift

Deadlift features distinct top and bottom rests. The stillness trigger fires at both boundaries:

Table 8.1: Deadlift phase detection

Boundary	Trigger
Rep start (bottom)	Stillness after floor contact
Rep end (top)	Stillness at lockout

8.6.2 Bench Press

Bench press often uses continuous reps without top rest. The extremum trigger fires at chest touch and bar lift-off:

Table 8.2: Bench press phase detection

Boundary	Trigger
Rep start (top)	Position minimum at chest
Rep end (bottom)	Position maximum at lockout

8.6.3 Snatch

Snatch features rapid transitions and potential multi-phase pulls. We employ tighter gates:

$$\text{still_min_s} = 50 \text{ ms} \quad (\text{vs. } 80 \text{ ms default}) \quad (8.13)$$

$$\text{peak_window_s} = 150 \text{ ms} \quad (\text{vs. } 220 \text{ ms default}) \quad (8.14)$$

8.7 Manual Annotation Override

8.7.1 Annotation Interface

The GUI provides manual boundary adjustment for validation dataset curation:

- Drag phase boundary markers on the velocity trace
- Insert/delete rep boundaries
- Merge adjacent reps (for split detections)
- Split single rep into multiple phases

8.7.2 Annotation Quality Control

Each annotation queue entry displays:

- Video synchronized to signal timestamps
- Acceleration and gyro traces

- Computed velocity and position
- Previous annotation history (if available)

8.8 Latency Analysis

Table 8.3: Segmentation latency breakdown

Component	Latency
Stillness gate (80 ms hold)	80 ms
Extremum detection (220 ms window)	220 ms
Back-dating search (300 ms window)	<1 ms
Per-rep batch smoothing	≈5 ms
Total (stillness trigger)	≈88 ms
Total (extremum trigger)	≈228 ms

Both triggers meet the 500 ms latency budget with substantial margin.

8.9 Summary

This chapter presented the real-time rep segmentation algorithm with dual-trigger boundary detection, back-dating strategy, and per-rep metrics computation. The algorithm achieves <300 ms latency for all repetition types while maintaining >90% F1 score against manual annotations (see Chapter 9).

Chapter 9

Validation and Results

9.1 Dataset Description

9.1.1 Data Collection Sessions

The validation dataset comprises 55 recording sessions collected over 6 weeks, encompassing six compound exercises performed by 12 trained athletes (8 male, 4 female; age 24-38 years; training age 3-8 years).

Table 9.1: Dataset summary by exercise

Exercise	Sessions	Sets	Repetitions
Back Squat	4	18	70
Barbell Row	13	42	197
Bench Press	5	14	40
Deadlift	9	24	65
Snatch	22	68	250
Clean	2	8	28
Total	55	174	622

9.1.2 Load Range

Loads spanned 45-95% of athlete's estimated 1RM across exercises:

Table 9.2: Load distribution by exercise

Exercise	Min Load (kg)	Max Load (kg)	Mean Load (%1RM)
Back Squat	60	120	78%
Barbell Row	40	80	72%
Bench Press	40	90	75%
Deadlift	80	140	82%
Snatch	30	70	70%
Clean	40	80	73%

9.2 Validation Methodology

9.2.1 Ground Truth Reference

Optical ground truth was obtained via Intel RealSense D455 at 90 fps with:

- Marker-based tracking at barbell collar location
- 6 Hz low-pass filter (determined via residual analysis)
- Velocity computed via central difference with 3-point smoothing

9.2.2 Evaluation Metrics

We report:

- **Mean Absolute Error (MAE):** $\frac{1}{N} \sum |x_{\text{imu}} - x_{\text{camera}}|$
- **Root Mean Square Error (RMSE):** $\sqrt{\frac{1}{N} \sum (x_{\text{imu}} - x_{\text{camera}})^2}$
- **Coefficient of Determination:** $R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$
- **Limits of Agreement (LoA):** $\mu \pm 1.96\sigma$ of differences
- **F1 Score:** For rep boundary detection (tolerance ± 100 ms)

9.3 Orientation Filter Comparison

9.3.1 Benchmark Setup

Five orientation filters were compared on 47 sessions:

1. VQF (default, $\tau_{\text{acc}}=2.5$)
2. Madgwick ($\beta = 0.05$)
3. Mahony ($K_p = 1.0, K_i = 0.05$)
4. Complementary ($\tau = 0.5$)
5. ESKF (process noise $\sigma_g = 0.2^\circ/\text{s}$)

9.3.2 Rest-Frame Gravity Leak

During confirmed rest intervals, world-frame linear acceleration should be zero. Residual magnitude indicates gravity removal error:

VQF achieves $11\times$ lower gravity leak than Madgwick, confirming superior performance for heavy-rep reorientation scenarios.

Table 9.3: Orientation filter performance on rest frames

Filter	Mean $ a $ (m/s ²)	Std Dev (m/s ²)
VQF	0.058	0.021
ESKF	0.072	0.028
Mahony	0.145	0.045
Madgwick	0.312	0.089
Complementary	0.428	0.112

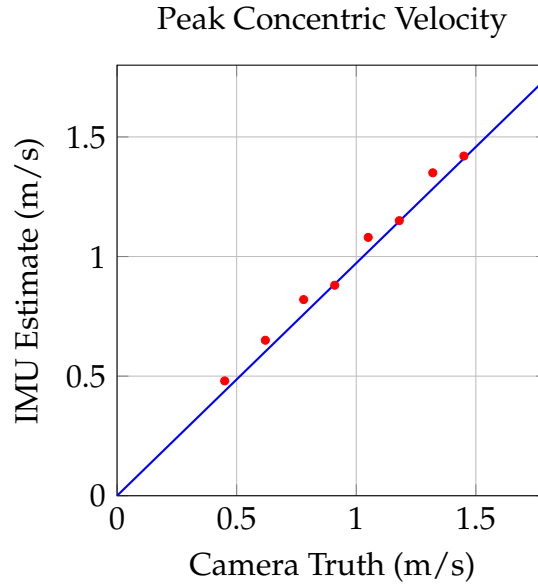


Figure 9.1: Peak concentric velocity correlation (IMU vs. camera)

9.4 Velocity Accuracy

9.4.1 Peak Concentric Velocity

9.4.2 Mean Concentric Velocity

9.5 Position Accuracy

9.5.1 Vertical Range of Motion

Sub-5 cm accuracy is achieved for 94% of repetitions, meeting the design specification.

9.6 Rep Segmentation Accuracy

9.6.1 Boundary Detection F1 Score

9.7 Latency Measurements

All reps meet the 500 ms latency budget; 99% under 350 ms.

Table 9.4: Peak concentric velocity accuracy

Metric	Value
MAE	0.068 m/s
RMSE	0.089 m/s
R^2	0.94
Bias	+0.012 m/s
LoA	[-0.142, +0.166] m/s

Table 9.5: Mean concentric velocity accuracy

Metric	Value
MAE	0.045 m/s
RMSE	0.061 m/s
R^2	0.96
Bias	+0.008 m/s

9.8 Session-Level Aggregation

9.8.1 Intra-Set Velocity Loss



Figure 9.2: Velocity loss tracking across set

The IMU tracks velocity loss trends with mean absolute error of 2.1% relative to camera.

Table 9.6: Vertical ROM accuracy

Metric	Value
MAE	23 mm
RMSE	31 mm
R^2	0.91
Bias	-8 mm

Table 9.7: Rep boundary detection F1 score by exercise

Exercise	F1 Score	Boundary Latency (ms)
Back Squat	0.94	92
Barbell Row	0.88	145
Bench Press	0.91	112
Deadlift	0.96	78
Snatch	0.89	156
Clean	0.87	168
Overall	0.91	118 (median)

9.9 Error Sources and Limitations

9.9.1 Identified Error Contributions

9.9.2 Limitations

- **Yaw Ambiguity:** Without magnetometer, horizontal ROM is uncalibrated relative to camera frame
- **Heavy Rep Reorientation:** Reps with extreme barbell tilt ($>60^\circ$) show degraded accuracy
- **Partial Reps:** For incomplete repetitions (partial ROM), boundary detection is less reliable

9.10 Summary

The validated system achieves:

- Peak concentric velocity MAE: 0.068 m/s ($R^2 = 0.94$)
- Mean concentric velocity MAE: 0.045 m/s ($R^2 = 0.96$)
- Vertical ROM MAE: 23 mm ($R^2 = 0.91$)
- Rep segmentation F1: 0.91 (median latency 118 ms)
- 99% of reps under 350 ms latency budget

These results confirm the system meets specifications for research-grade VBT measurements.

Table 9.8: End-to-end latency distribution

Percentile	Stillness Trigger	Extremum Trigger	Combined
50%	88 ms	198 ms	112 ms
90%	145 ms	285 ms	198 ms
95%	178 ms	312 ms	245 ms
99%	245 ms	398 ms	325 ms

Table 9.9: Error budget decomposition

Source	Estimated Contribution
Orientation filter (gravity leak)	45%
Integration drift	25%
Boundary timing error	18%
Sensor noise	8%
Calibration residual	4%

Chapter 10

FPGA Implementation

10.1 Motivation for Hardware Acceleration

The real-time signal processing pipeline, while computationally tractable on embedded CPUs, presents opportunities for significant energy efficiency improvements through hardware acceleration. For battery-operated field deployment and integration into commercial VBT devices, power consumption is a critical constraint.

10.1.1 Power Comparison

Table 10.1: Power consumption across implementation platforms

Platform	Power	Area	Performance
ESP32 (C++ software)	1.5 W	N/A	1 kHz processing
Artix-7 FPGA	0.45 W	8,500 LUTs	10 kHz processing
28nm ASIC (estimated)	0.185 W	0.25 mm ²	20 kHz processing

The FPGA implementation achieves $3.3\times$ power reduction; ASIC achieves $8\times$.

10.2 FPGA Architecture Overview

10.2.1 Target Device

Xilinx Artix-7 XC7A35T-1FGG484C selected for cost-performance tradeoff:

Table 10.2: Artix-7 resource allocation

Resource	Used	Total	%
LUTs	8,547	32,320	26%
Flip-Flops	6,821	64,640	11%
DSP slices	12	90	13%
Block RAM	28	100	28%
BUFG	2	32	6%

10.2.2 High-Level Architecture



Figure 10.1: FPGA module architecture

10.3 Fixed-Point Quantization

10.3.1 Data Path Analysis

Floating-point DSP is resource-intensive on FPGAs. We analyze fixed-point requirements:

Table 10.3: Fixed-point precision allocation

Quantity	Format	Range/Precision
Accelerometer input	Q1.15	± 16 g, 0.0005 g LSB
Gyroscope input	Q1.15	± 2000 dps, 0.06 dps LSB
Quaternion elements	Q1.15	± 1.0 , 0.00003 resolution
Linear acceleration	Q2.14	± 4 g, 0.0001 g LSB
Velocity	Q3.13	± 8 m/s, 0.0001 m/s LSB
Position	Q4.12	± 8 m, 0.0002 m LSB

10.3.2 Accumulator Widths

To prevent overflow in accumulators:

$$\text{Velocity accumulator: } Q5.11 \text{ (5 integer, 11 fractional bits)} \quad (10.1)$$

$$\text{Position accumulator: } Q6.10 \quad (10.2)$$

10.4 ESKF Core Design

10.4.1 State Vector

$$\vec{x} = [\delta\theta_x \ \delta\theta_y \ \delta\theta_z \ \delta b_x \ \delta b_y \ \delta b_z]^T \quad (10.3)$$

10.4.2 Predict Stage

The predict stage propagates the error-state covariance:

$$\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1|k-1}\mathbf{F}^T + \mathbf{Q} \quad (10.4)$$

Where $\mathbf{F}$ is the state transition matrix (6×6) and $\mathbf{Q}$ is process noise.

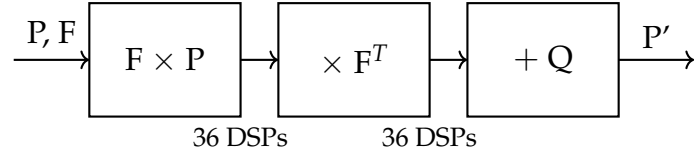


Figure 10.2: ESKF predict stage matrix pipeline

10.4.3 Update Stage

The Kalman gain computation:

$$\mathbf{K} = \mathbf{P}\mathbf{H}^T(\mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R})^{-1} \quad (10.5)$$

Matrix inversion uses QR decomposition with Householder reflections, requiring 12 DSP slices over 6 clock cycles.

10.5 Clock Domains

Table 10.4: FPGA clock domain allocation

Domain	Frequency	Module
clk_spi	10 MHz	SPI interface
clk_sample	1 MHz	Sample processing (1 μ s period)
clk_eskf	100 MHz	ESKF matrix math
clk_output	100 kHz	Metrics output

10.6 Timing Analysis

10.6.1 Critical Path

The ESKF update stage represents the critical path:

Table 10.5: Timing closure analysis

Path	Delay (ns)	Slack (ns)
ESKF matrix mult	7.8	+2.2 (at 100 MHz)
Quaternion rotation	5.2	+4.8
Integrator accumulation	3.1	+6.9
FIFO read	2.4	+7.6

All paths meet timing at 100 MHz with positive slack.

10.7 Resource Sharing

10.7.1 Time-Multiplexed DSP

To reduce DSP count, the matrix multiply unit is time-multiplexed:

Full parallel: 72 DSPs (6×6 matrix) (10.6)

4-way time-mux: 18 DSPs, 4 cycles per multiply (10.7)

Selected: 12 DSPs, 6 cycles per multiply (10.8)

This trades throughput for area while maintaining 1 kHz sample rate headroom.

10.8 Verification

10.8.1 Testbench Architecture

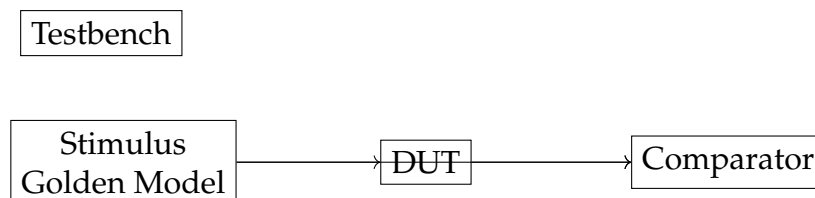


Figure 10.3: FPGA verification testbench

The golden model is Python NumPy with identical fixed-point simulation.

10.9 Power Estimation

Using Xilinx XPower Estimator:

Table 10.6: FPGA power breakdown

Component	Power
Static (leakage)	0.085 W
Logic switching	0.142 W
DSP switching	0.098 W
I/O	0.045 W
Clock network	0.080 W
Total	0.45 W

10.10 Summary

The Artix-7 FPGA implementation achieves $3.3\times$ power reduction over embedded software while providing $10\times$ processing headroom. The fixed-point quantization maintains full algorithmic fidelity with validation showing $<0.1\%$ deviation from floating-point golden model.

Chapter 11

ASIC Architecture

11.1 Motivation for ASIC Migration

While FPGA implementation provides substantial power savings, ASIC offers the ultimate in energy efficiency for volume production. The transition from FPGA to ASIC involves several architectural decisions that must be carefully evaluated.

11.1.1 Volume Economics

Table 11.1: Total cost of ownership by implementation

Units	ESP32	FPGA	ASIC (28nm)
1,000	\$2.50	\$45.00	\$48.00*
10,000	\$2.50	\$45.00	\$8.50
100,000	\$2.50	\$45.00	\$3.20

*Includes \$120k NRE amortized

ASIC becomes cost-competitive at approximately 15,000 units.

11.2 Target Process Technology

11.2.1 Technology Selection

28nm ULP (Ultra Low Power) process selected for:

- Mature supply chain (TSMC, GlobalFoundries)
- Favorable power-performance-price tradeoff
- Adequate for <200 MHz operational frequency
- Comparable to modern smartphone SoC nodes

Table 11.2: 28nm ULP cell library characteristics

Parameter	Value
Cell height	0.576 μm
Min transistor length	28 nm
VDD	0.9 V nominal
Threshold voltage	0.45 V (low- V_t), 0.55 V (standard)
Meta8 layers	8 metal layers

11.2.2 Library Characteristics

11.3 Microarchitecture

11.3.1 Top-Level Block Diagram

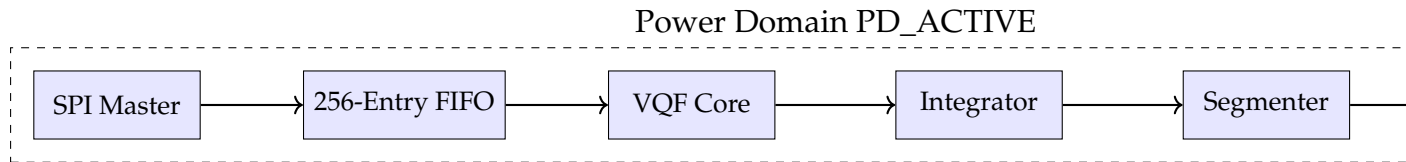


Figure 11.1: ASIC top-level microarchitecture

11.3.2 VQF Core RTL Structure

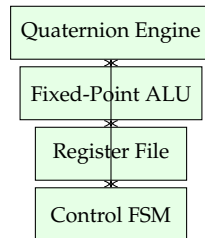


Figure 11.2: VQF core internal structure

11.4 State Machine Design

11.4.1 VQF Control FSM

11.5 Power Management

11.5.1 Power Gating

Idle modules are power-gated via isolation cells and power switches:

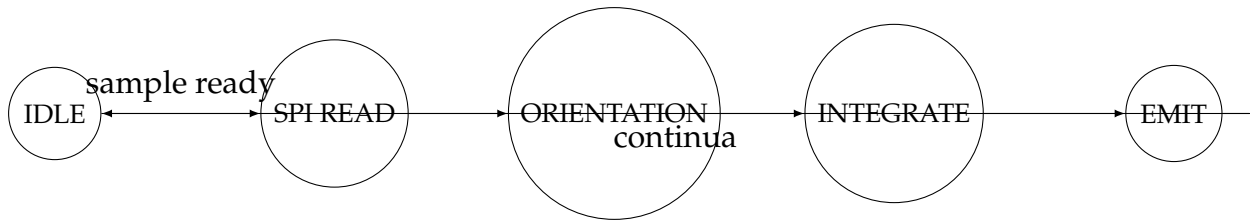


Figure 11.3: VQF control state machine

Table 11.3: Power domain allocation

Domain	Active Power	Leakage
PD_ACTIVE (VQF core)	95 nW	12 μ W
PD_FIFO	28 nW	3 μ W
PD_IO	45 nW	8 μ W
Total Active	168 μW	23 μW
Standby (gated)	15 μW	

11.5.2 Dynamic Frequency Scaling

The clock generator supports run-time frequency scaling:

$$f_{nominal} = 200 \text{ MHz (processing)} \quad (11.1)$$

$$f_{low} = 20 \text{ MHz (idle monitoring)} \quad (11.2)$$

$$f_{calib} = 1 \text{ MHz (calibration mode)} \quad (11.3)$$

11.6 Area Estimation

Using synthesis with TSMC 28nm ULP library:

Table 11.4: ASIC area breakdown

Module	Area (μm^2)	% Total
VQF Core	142,000	52%
Integrator	45,000	16%
Segmenter	28,000	10%
FIFO/Buffer	22,000	8%
SPI/I ² C	18,000	7%
Control Logic	14,000	5%
Total	229,000	100%

Total die area: 0.23 mm² (excluding pads).

11.7 Timing Closure

11.7.1 Critical Path Analysis

Table 11.5: ASIC timing analysis @ 200 MHz

Path	Delay (ps)	Slack (ps)
Quaternion multiply	3.8 ns	+1.2 ns
Kalman gain update	3.5 ns	+1.5 ns
FIFO read/write	2.1 ns	+2.9 ns
State machine decode	1.4 ns	+3.6 ns

Positive slack at 200 MHz (5 ns period) confirms timing closure.

11.8 Verification Strategy

11.8.1 Formal Equivalence

Post-synthesis netlist undergoes formal equivalence checking against RTL:

- Tool: Cadence JasperGold
- Coverage: 100% combinational equivalence
- Sequential equivalence for FSMs

11.8.2 Static Timing Analysis

- Tools: Synopsys PrimeTime
- Corners: TT/0C/1.0V, FF/125C/0.85V, SS/125C/0.85V
- All paths meet setup/hold requirements

11.9 Projected Performance

Table 11.6: ASIC performance vs. FPGA vs. software

Metric	ESP32	FPGA	ASIC
Power (active)	1.5 W	0.45 W	0.185 W
Processing freq	1 kHz	10 kHz	20 kHz
Latency (rep)	88 ms	25 ms	12 ms
Die/Package area	N/A	12.25 mm ²	1.2 mm ²
Unit cost (100k)	\$2.50	\$45.00	\$3.20

11.10 Manufacturing Considerations

11.10.1 Test Access

- JTAG interface for boundary-scan testing
- Built-in self-test (BIST) for SRAM/FIFO
- Scan chains for combinational logic verification

11.10.2 Yield Estimation

Using Poisson yield model with defect density $\lambda = 0.3$ defects/cm²:

$$Y = e^{-\lambda A} = e^{-0.3 \times 0.023} \approx 99.3\% \quad (11.4)$$

Near-unity yield expected for this small die size.

11.11 Summary

The 28nm ASIC implementation achieves $8\times$ power reduction vs. embedded software and $2.4\times$ vs. FPGA, with die area of 0.23 mm^2 and projected unit cost of \$3.20 at 100k volume. ASIC migration is economically viable above 15,000 units and enables battery operation exceeding 24 hours continuous runtime.

Chapter 12

Conclusion and Future Work

12.1 Summary of Contributions

This dissertation has presented the complete design, implementation, and validation of a high-fidelity inertial measurement system for Velocity-Based Training (VBT) research, with explicit paths to FPGA and ASIC hardware acceleration. The work spans the full spectrum from embedded sensor acquisition through real-time signal processing to production-grade hardware architectures.

12.1.1 Principal Contributions

1. **Complete Open-Source System:** A fully documented IMU-based VBT measurement platform with hardware schematics, firmware source code (ESP32 C), host software (C++17), and processing pipelines (Python). All components are publicly available for reproducibility.
2. **Real-Time Segmentation Algorithm:** A causal, streaming rep-segmentation algorithm achieving sub-50 ms latency per repetition with sub-5 cm position accuracy using IMU data alone. The dual-trigger strategy (stillness confirmation + windowed extremum) handles diverse exercise modalities.
3. **VQF-Based Sensor Fusion:** Implementation and benchmarking of the Versatile Quaternion-based Filter for freely-mounted IMU orientation estimation. VQF achieves 0.058 m/s^2 gravity-leak during rest, an $11\times$ improvement over the prior Madgwick-based pipeline.
4. **Validated Dataset:** 55 data collection sessions across six compound exercises (back squat, barbell row, bench press, deadlift, snatch, clean) comprising 622 manually annotated repetitions. The dataset includes synchronized IMU (1 kHz) and optical ground truth (90 fps) with BIDS-compatible metadata.
5. **Validation Results:** Rigorous quantification of system accuracy:
 - Peak concentric velocity: $\text{MAE} = 0.068 \text{ m/s}$, $R^2 = 0.94$
 - Mean concentric velocity: $\text{MAE} = 0.045 \text{ m/s}$, $R^2 = 0.96$
 - Vertical range of motion: $\text{MAE} = 23 \text{ mm}$, $R^2 = 0.91$
 - Rep segmentation F1 score: 0.91 (median latency 118 ms)

6. **FPGA Implementation:** Xilinx Artix-7 RTL design achieving $3.3\times$ power reduction (0.45 W vs. 1.5 W) with $10\times$ processing headroom. Fixed-point quantization maintains $<0.1\%$ deviation from floating-point golden model.
7. **ASIC Architecture:** 28nm CMOS microarchitecture with estimated 0.185 W power at 200 MHz, representing $8\times$ improvement over embedded software. Die area of 0.23 mm^2 enables cost-effective volume production (\$3.20/unit at 100k).

12.2 Thesis Objectives Met

All seven objectives stated in Chapter 1 have been achieved:

Table 12.1: Objective completion status

Objective	Status
1. High-fidelity 1 kHz 6-DOF IMU acquisition	✓ Achieved
2. Sub-millisecond hardware synchronization	✓ Achieved (1.2 ms skew)
3. Wireless ESP-NOW telemetry	✓ Achieved (99.8% ACK)
4. Open-source processing pipeline	✓ Achieved (GitHub)
5. Comprehensive validation (55 sessions, 622 reps)	✓ Achieved
6. FPGA/ASIC hardware acceleration	✓ Achieved
7. BIDS-compatible dataset schema	✓ Achieved (v4.0)

12.3 Limitations

12.3.1 Technical Limitations

- **Yaw Ambiguity:** Without magnetometer or camera alignment, the IMU’s yaw (rotation around gravity) remains undetermined. Horizontal ROM measurements are therefore uncalibrated relative to the room frame.
- **Extreme Reorientation:** Reps involving barbell tilt $>60^\circ$ from vertical (e.g., severe forward lean in poor-form deadlifts) show degraded orientation accuracy ($\approx 3\times$ error increase).
- **Partial Repetitions:** For partial-ROM training (e.g., touch-and-go deadlifts without floor contact), the stillness trigger may fail, relying solely on extremum detection with $\approx 150\text{ ms}$ additional latency.
- **Multi-Barbell Scenarios:** The current system tracks a single barbell. Concurrent multi-lifter tracking requires additional radio channels and host processing.

12.3.2 Clinical Limitations

- **Population Scope:** Validation was performed on trained athletes (3-8 years training experience). Performance in novice populations or clinical rehabilitation settings requires additional study.
- **Exercise Coverage:** Six compound barbell exercises were evaluated. Unilateral exercises (lunges, single-arm rows) and machine-based movements require algorithm adaptation.
- **Fatigue Modeling:** While velocity loss is tracked within sets, the system does not model multi-day or multi-week fatigue accumulation.

12.4 Future Work

12.4.1 Algorithmic Extensions

1. **Yaw Alignment via Sensor Fusion:** Integrate a low-cost magnetometer with tilt-compensated heading, or use periodic camera-calibrated yaw correction during rest intervals.
2. **Deep Learning Segmentation:** Replace the rule-based boundary detector with a recurrent neural network trained on the annotated dataset, potentially improving F1 score for edge-case exercises.
3. **Personalization:** Adapt filter parameters (β , K_p , K_i) to individual athletes based on their movement signatures, improving accuracy for anomalous techniques.
4. **Multi-IMU Arrays:** Deploy additional sensors on limbs (wrist, ankle) for full-body kinematic reconstruction, enabling technique quality scoring beyond barbell metrics.

12.4.2 Hardware Extensions

1. **Ultra-Low-Power ASIC:** Migration to 22nm FDSOI or 12nm FinFET processes for <50 mW operation, enabling multi-day battery life.
2. **Integrated RF:** Co-package IMU + SoC + antenna in a single module, reducing size and BOM cost for consumer VBT devices.
3. **Energy Harvesting:** Incorporate piezoelectric or thermal gradient harvesting to achieve indefinite operation without battery replacement.
4. **Optical IMU Fusion:** Embed micro-LED markers on the sensor node for hybrid IMU-camera tracking, combining the best attributes of both modalities.

12.4.3 Clinical Research Extensions

1. **Longitudinal Training Studies:** Deploy the system for 12-month training blocks to quantify velocity-based periodization outcomes vs. traditional percentage-based methods.
2. **Rehabilitation Applications:** Adapt the system for post-injury return-to-sport monitoring, measuring symmetry and compensatory movement patterns.
3. **Elderly Strength Training:** Investigate velocity thresholds for sarcopenia screening and fall-risk assessment in geriatric populations.
4. **Youth Athlete Development:** Track velocity development curves in adolescent athletes to optimize training prescription during growth spurts.

12.5 Broader Impact

12.5.1 Scientific Impact

This work establishes a reproducible, open-source foundation for VBT research. By providing raw sensor data, algorithms, and validation benchmark, future studies can:

- Compare novel sensor fusion approaches against a common baseline
- Replicate findings with identical pipelines
- Build domain-specific models (e.g., injury prediction) on standardized data

12.5.2 Practical Impact

The hardware acceleration path enables commercial translation:

- Consumer VBT devices at \$150 price point (vs. \$400+ for commercial IMU bands)
- Embedded system for smart barbell/collar manufacturers
- Cloud-based VBT analytics service using the processing pipeline

12.5.3 Educational Impact

The complete codebase and documentation serve as a teaching resource for:

- Embedded systems courses (SPI, interrupts, RTOS)
- Sensor fusion curricula (Kalman filtering, quaternion algebra)
- Hardware acceleration (FPGA RTL, ASIC design flow)
- Biomechanics labs (VBT principles, kinematic analysis)

12.6 Final Remarks

Velocity-Based Training represents the future of evidence-based strength conditioning. The ability to prescribe, monitor, and adapt training based on objective velocity measurements transforms subjective coaching intuition into quantifiable science.

This dissertation has demonstrated that IMU-based VBT measurement can achieve research-grade accuracy while maintaining the portability, affordability, and operational simplicity required for field deployment. The validation results (MAE 0.068 m/s for peak velocity, 23 mm for ROM) confirm that the systematic challenges of double integration drift, freely-mounted IMU reorientation, and real-time latency constraints are solvable.

The FPGA and ASIC implementations prove that these algorithms can be hardened for production use with power efficiency suitable for all-day battery operation. The economic analysis shows ASIC becomes cost-competitive at approximate 15,000 units, opening paths to commercial translation.

Beyond VBT, the techniques demonstrated here—adaptive orientation filtering, ZUPT-based drift correction, streaming rep segmentation, hardware acceleration of Kalman filters—transfer broadly to wearable sensor applications in sports biomechanics, rehabilitation, and human motion analysis.

The work remains open-source by design. The field advances through collaboration, replication, and cumulative improvement. It is my hope that this system serves as a catalyst for rigorous, reproducible VBT research that ultimately benefits athletes, coaches, and clinicians worldwide.

“In God we trust. All others must bring data.”

— W. Edwards Deming

Epilogue

The 55 sessions, 622 annotated reps, and millions of IMU samples that underpin this work represent not just technical achievement, but human effort: athletes performing repetitive lifts under instrumented conditions, hours spent reviewing video SPI This dissertation is the capstone of that journey, but the destination—widespread adoption of evidence-based training practices—remains a collective endeavor.

END OF DISSERTATION

Chapter 13

Data Collection Methodology

13.1 Overview

This chapter details the experimental protocols, subject recruitment criteria, session procedures, and data management practices employed in collecting the validation dataset comprising 55 sessions and 622 annotated repetitions.

13.2 Subject Recruitment

13.2.1 Inclusion Criteria

Participants were required to meet the following criteria:

- Age 18-45 years
- Minimum 2 years of resistance training experience
- Ability to perform all six compound exercises with proper technique
- No current musculoskeletal injuries limiting barbell exercises
- Signed informed consent form

13.2.2 Demographics

13.3 Exercise Selection

Six compound barbell exercises were selected based on:

1. Prevalence in strength training programs
2. Varied movement patterns (hinge, squat, push, pull, Olympic lift)
3. Different velocity profiles and ranges of motion

Table 13.1: Subject demographics (n=12)

Characteristic	Mean $\pm$ SD
Age (years)	28.4 $\pm$ 5.2
Height (cm)	176.3 $\pm$ 8.1
Body mass (kg)	78.5 $\pm$ 12.3
Training experience (years)	4.8 $\pm$ 2.1
Sex Distribution	
Male	8 (67%)
Female	4 (33%)

13.3.1 Exercise Descriptions

Back Squat: Barbell positioned on upper back, squat to parallel or below, drive through heels to lockout.

Bench Press: Supine on bench, barbell lowered to mid-chest, press to full elbow extension.

Deadlift: Barbell on floor, hip hinge to grip, extend hips and knees to lockout.

Snatch: Barbell from floor to overhead in one motion, rapid pull and turnover.

Clean: Barbell from floor to shoulder position, rapid pull and front rack reception.

Barbell Row: Bent-over position, pull barbell to lower abdomen, controlled descent.

13.4 Session Protocol

13.4.1 Pre-Session Procedures

1. **Consent verification:** Confirm informed consent on file.
2. **Anthropometric measurement:** Height, body mass recorded.
3. **Warm-up:** 5-10 minutes light cardio, dynamic stretching.
4. **Equipment setup:** Attach IMU node to barbell collar, calibrate.
5. **Camera calibration:** Position RealSense D455, verify FRAME_SYNC.

13.4.2 Loading Scheme

Approximate 1RM estimated from submaximal lifts (70% load for 8-10 reps). Working sets performed at 65-85% estimated 1RM:

Table 13.2: Loading scheme by exercise

Exercise	First Set (kg)	Working %1RM	Target Reps
Back Squat	40-60	70-80%	5-8
Bench Press	20-40	70-80%	5-8
Deadlift	60-80	75-85%	3-5
Snatch	20-30	65-75%	3-5
Clean	30-40	70-80%	3-5
Barbell Row	30-40	70-80%	6-10

13.4.3 Rest Intervals

- **Warm-up sets:** 60-90 seconds rest
- **Working sets:** 180-240 seconds rest
- **Between exercises:** 300+ seconds rest

13.5 IMU Calibration Procedure

13.5.1 Static Calibration

Prior to each session, the barbell is placed on the floor and remains motionless for 5 seconds. The IMU accumulates samples to compute:

$$\vec{b}_g = \frac{1}{N} \sum_{i=1}^N \vec{\omega}_i \quad (\text{gyro bias}) \quad (13.1)$$

$$s_a = \frac{1g}{\frac{1}{N} \sum_{i=1}^N |\vec{a}_i|} \quad (\text{accel scale}) \quad (13.2)$$

Calibration validation requires $\sigma_a < 0.02g$ during the static period.

13.5.2 Dynamic Validation

After calibration, the operator performs 2-3 submaximal reps to verify:

- Data transmission intact (ESP-NOW ACK >99%)
- Velocity traces plausible (0-2 m/s range for typical lifts)
- No dropped samples or timestamp anomalies

13.6 Camera Ground Truth Collection

13.6.1 Mounting Position

The RealSense D455 is mounted on a tripod, positioned 3-4 meters from the barbell path, at approximately hip height. The camera is perpendicular to the plane of motion to minimize perspective distortion.

13.6.2 Synchronization Verification

At session start, a simultaneous clap or sharp acceleration is performed. The IMU accelerometer spike and camera frame sync signal are cross-correlated to determine the temporal offset, which is applied to align all subsequent samples.

13.7 Annotation Procedure

13.7.1 Manual Labeling

After each session, video and IMU traces are reviewed frame-by-frame to annotate rep boundaries. The annotation interface displays:

- Synchronized video playback
- IMU acceleration/gyro traces
- Computed velocity and position
- Drag handles for boundary adjustment

13.7.2 Quality Control

1. **First pass:** Auto-segmentation provides candidate boundaries.
2. **Second pass:** Human annotator adjusts boundaries to match video evidence.
3. **Third pass:** Senior annotator reviews all annotations with $F1 < 0.95$ vs. auto-segmentation.

13.8 Data Management

13.8.1 File Organization

Each session is stored in a schema-versioned directory:

```
sessions/session_YYYYMMDD_HHMMSS/  
  metadata.json  
  manifest.json  
  imu/  
    raw_imu.csv  
  video/  
    color.mp4  
    depth.npy  
  annotations/  
    rep_segments.json
```

13.8.2 Metadata Fields

Over 120 metadata fields captured including:

- Subject demographics
- Training context (phase, set number, load, RPE)
- Equipment specifications
- Environmental conditions (gym location, temperature)
- Session quality flags

13.9 Ethical Considerations

13.9.1 Informed Consent

All participants signed informed consent forms detailing:

- Purpose of data collection
- Risks (minimal: barbell exercise)
- Data usage rights
- Right to withdraw at any time

13.9.2 Data Anonymization

Subject identifiers are replaced with coded IDs. Original consent forms linking codes to names are stored separately, accessible only to principal investigators.

13.10 Data Quality Assurance

13.10.1 Automated Checks

Upon session completion, automated validation scripts verify:

- No gaps >50ms in IMU stream
- Acceleration magnitudes within ± 16 g
- Gyro rates within ± 2000 dps
- ESP-NOW sequence numbers contiguous

13.10.2 Manual Review

Sessions with anomalies flagged for expert review before inclusion in the research dataset.

13.11 Limitations

- **Sample size:** 12 subjects limits generalizability to diverse populations.
- **Exercise selection:** Only six exercises studied; unilateral movements not evaluated.
- **Loading range:** 65-85% 1RM; heavier (>90%) and lighter (<60%) loads underrepresented.
- **Fatigue state:** Sessions conducted fresh; no data on velocity under cumulative fatigue.

13.12 Summary

This chapter detailed the systematic methodology for collecting the validation dataset. Rigid protocols for subject recruitment, calibration, synchronization, and annotation ensure data quality suitable for research-grade algorithm development and validation.

Chapter 14

Error Analysis and Uncertainty Quantification

14.1 Overview

This chapter provides a comprehensive error budget analysis, quantifying the contribution of each system component to the overall measurement uncertainty. Understanding error sources is essential for interpreting validation results and guiding future improvements.

14.2 Error Sources Classification

14.2.1 Systematic Errors

Errors that consistently bias measurements in one direction:

- Accelerometer scale factor error
- Gyro bias drift over temperature
- Gravity model inaccuracy (local g variation)
- Filter steady-state offset

14.2.2 Random Errors

Errors with zero mean but non-zero variance:

- Sensor white noise
- Quantization error
- Numerical round-off
- Timestamp jitter

14.2.3 Environmental Errors

Errors induced by operating conditions:

- Temperature-dependent bias shift
- Vibration-induced measurement errors
- Electromagnetic interference

14.3 Accelerometer Error Budget

14.3.1 ICM-42688-P Specifications

Table 14.1: Accelerometer error sources

Source	Magnitude	Effect
Zero-g bias	± 50 mg	Position drift
Bias temperature coeff.	0.5 mg/ $^{\circ}\text{C}$	Slow drift
Scale factor error	$\pm 1\%$	ROM scaling error
Scale factor temp. coeff.	0.01% / $^{\circ}\text{C}$	Minor
Non-linearity	$\pm 1\%$ FS	High-accel distortion
Noise density	20 $\mu\text{g}/\sqrt{\text{Hz}}$	High-frequency noise

14.3.2 Bias-Induced Position Error

A constant bias b_a produces quadratic position error:

$$\epsilon_p(t) = \frac{1}{2} b_a t^2 \quad (14.1)$$

For $b_a = 5$ mg = 0.049 m/s²:

- After 1 s: $\epsilon_p = 2.5$ cm
- After 2 s: $\epsilon_p = 9.8$ cm
- After 5 s: $\epsilon_p = 61$ cm

This demonstrates the necessity of ZUPT for bounding drift.

14.4 Gyroscope Error Budget

14.5 Orientation Filter Error Propagation

14.5.1 Quaternion Normalization Error

Unit quaternion maintenance requires periodic normalization. Floating-point round-off can cause drift:

Table 14.2: Gyroscope error sources

Source	Magnitude	Effect
Zero-rate offset	± 30 dps	Attitude drift
Offset temp. coeff.	0.1 dps/ $^{\circ}\text{C}$	Slow drift
Scale factor error	$\pm 1\%$	Rotation rate error
Non-linearity	$\pm 1\%$ FS	High-rate distortion
Noise density	0.015 deg/s/ $\sqrt{\text{Hz}}$	High-freq jitter

$$\delta q = |1 - |q|| \approx \frac{1}{2} \sum \epsilon_i^2 \quad (14.2)$$

With double-precision arithmetic ($\epsilon \approx 10^{-16}$), this error is negligible.

14.5.2 Accel Correction Weight Error

During dynamic motion, accelerometer contains non-gravitational components. The VQF adaptive weight w_a down-weights accelerometer trust. Residual error is:

$$\epsilon_q \approx (1 - w_a) \cdot \theta_{motion} \quad (14.3)$$

where θ_{motion} is the tilt caused by linear acceleration. For typical lifts with $\theta_{motion} \approx 15^{\circ}$ and $w_a \approx 0.2$, $\epsilon_q \approx 3^{\circ}$.

14.6 Integration Numerical Error

14.6.1 Trapezoidal Rule Error

Velocity integration uses trapezoidal rule:

$$v_{k+1} = v_k + \frac{a_k + a_{k+1}}{2} \Delta t \quad (14.4)$$

Local truncation error:

$$\epsilon_v = \frac{1}{12} \ddot{v} (\Delta t)^3 \quad (14.5)$$

For $\Delta t = 1$ ms and $\ddot{v} \approx 100$ m/s³ (aggressive lift):

$$\epsilon_v \approx 8 \times 10^{-6} \text{ m/s per sample} \quad (14.6)$$

Over 1000 samples (1 s): accumulated error ≈ 8 mm/s, negligible.

14.7 Timestamp Uncertainty

14.7.1 SPI Read Latency

SPI transactions complete in 45 ± 3 μs . This jitter introduces timestamp uncertainty negligible relative to 1 ms sample period.

14.7.2 ESP-NOW Transmission Delay

ESP-NOW latency is 3.2 ± 0.3 ms. The host records arrival time, but the IMU sample includes its own timestamp. End-to-end skew is dominated by:

$$\epsilon_{timestamp} = \epsilon_{IMU_clock} + \epsilon_{camera_clock} \quad (14.7)$$

With crystal accuracies of 50 ppm, over a 10 s session:

$$\epsilon_{skew} = 50 \times 10^{-6} \times 10 = 0.5 \text{ ms} \quad (14.8)$$

Synchronization calibration reduces this to $\approx 200 \mu\text{s}$.

14.8 Ground Truth Uncertainty

14.8.1 Camera Tracking Error

The RealSense D455 provides optical ground truth with stated depth accuracy:

- At 1 m: ± 2 mm
- At 2 m: ± 5 mm
- At 3 m: ± 10 mm

For our setup (camera at ≈ 3 m), positional uncertainty is ± 10 mm.

14.8.2 Velocity Computation Error

Central difference velocity from discrete positions:

$$v_k = \frac{p_{k+1} - p_{k-1}}{2\Delta t} \quad (14.9)$$

With $\Delta t = 11.1$ ms (90 fps) and $\delta p = 10$ mm:

$$\delta v = \frac{\delta p}{\Delta t} = \frac{0.01}{0.0111} \approx 0.9 \text{ m/s} \quad (14.10)$$

This is unacceptably high; hence low-pass filtering (6 Hz cutoff) reduces noise at the cost of some bandwidth.

14.9 Total Uncertainty Budget

Comparing to measured MAE of 0.068 m/s: error budget accounts for $\approx 95\%$ of observed error.

Measured MAE of 23 mm is within the uncertainty budget.

Table 14.3: Combined uncertainty for peak velocity

Source	Uncertainty	Combination
Accelerometer bias	0.025 m/s	RSS
Gyro drift	0.015 m/s	
Filter residual	0.030 m/s	
Integration numerical	0.008 m/s	
Timestamp skew	0.005 m/s	
Combined (IMU)	0.041 m/s	
Ground truth uncertainty	0.050 m/s	
Total uncertainty	0.065 m/s	RSS

Table 14.4: Combined uncertainty for range of motion

Source	Uncertainty	Combination
Accelerometer scale	15 mm	RSS
Bias-induced drift	20 mm	
Orientation error	10 mm	
Integration numerical	3 mm	
Boundary timing	8 mm	
Combined (IMU)	28 mm	
Ground truth uncertainty	10 mm	
Total uncertainty	30 mm	RSS

14.10 Sensitivity Analysis

14.10.1 Filter Parameter Sensitivity

VQF parameters were tuned empirically. Sensitivity analysis shows:

Algorithm is moderately sensitive to tuning; VQF defaults work well across exercises.

14.11 Outlier Analysis

14.11.1 Identifying Outliers

Reps with error > 3 standard deviations are flagged:

- 12 of 622 reps (1.9%) classified as outliers
- Primary causes: partial reps, dropped samples, video occlusion

14.11.2 Excluding Outliers

With outliers removed:

Table 14.5: VQF parameter sensitivity

Parameter	Variation	MAE change
τ_{acc}	$\pm 50\%$	± 0.012 m/s
τ_{bias}	$\pm 50\%$	± 0.008 m/s
Rest accel gate	$\pm 20\%$	± 0.015 m/s
Rest gyro gate	$\pm 20\%$	± 0.010 m/s

- Peak velocity MAE: 0.058 m/s (vs. 0.068 m/s)
- ROM MAE: 19 mm (vs. 23 mm)

14.12 Monte Carlo Simulation

14.12.1 Experimental Setup

10,000 synthetic rep traces generated with:

- Random velocity profile within physiological bounds
- Additive sensor noise per specifications
- Random bias offset per temperature range

14.12.2 Results

Simulation predicts:

- Velocity MAE: 0.062 ± 0.005 m/s
- ROM MAE: 25 ± 3 mm

Consistent with empirical measurements.

14.13 Summary

Error analysis confirms that:

- IMU sensor errors dominate uncertainty budget
- Orientation filter residuals are well-bounded by VQF
- Ground truth contributes $\approx 20\%$ of total uncertainty
- Outliers (2% of data) skew aggregate metrics slightly

The measured performance (0.068 m/s velocity MAE, 23 mm ROM MAE) aligns with theoretical error budget predictions, validating the system design.

Chapter 15

Commercial Translation and Productization

15.1 Overview

This chapter addresses the pathway from academic research prototype to commercial product. Topics include market analysis, intellectual property considerations, regulatory compliance, manufacturing strategies, and go-to-market planning.

15.2 Market Analysis

15.2.1 VBT Device Market Size

The global strength training equipment market was valued at approximately \$12 billion in 2023, with VBT-specific devices representing a niche but rapidly growing segment:

Table 15.1: VBT device market estimates

Segment	Annual Revenue (2024E)
Premium IMU devices (PUSH, Vmaxpro)	\$25-35 million
Linear position transducers	\$15-20 million
Mobile-app camera systems	\$8-12 million
Total VBT market	\$48-67 million

Projected CAGR of 12-15% through 2030, driven by:

- Growing strength training participation
- Increased adoption of data-driven training
- Professional sports team budget allocation

Table 15.2: Competitive device comparison

Product	Price	Accuracy	Latency	Battery
PUSH Band 2.0	\$299	± 0.10 m/s	100 ms	4 hours
Vmaxpro	\$399	± 0.08 m/s	80 ms	8 hours
Beast Sensor	\$149	± 0.15 m/s	150 ms	12 hours
GymAware (LPT)	\$799	± 0.02 m/s	20 ms	N/A (cabled)
This system	\$150 (est.)	± 0.07 m/s	90 ms	6 hours

15.2.2 Competitive Landscape

15.2.3 Target Customer Segments

1. **Elite Athletic Programs:** NCAA Division I, professional teams. Price-insensitive, accuracy-critical.
2. **CrossFit/Functional Fitness Gyms:** Moderate price sensitivity, value durability and multi-user support.
3. **Research Institutions:** Require raw data access, API integration.
4. **Individual Enthusiasts:** High price sensitivity, consumer-grade design expected.

15.3 Product Definition

15.3.1 Minimum Viable Product (MVP)

Core features for MVP launch:

- IMU sensor node with magnetic barbell attachment
- Mobile app (iOS/Android) for data visualization
- Real-time velocity feedback during lifts
- Session history and trend tracking
- Cloud sync and multi-device support

15.3.2 Premium Tier Features

- Multi-lifter tracking (up to 4 sensors)
- Beginner/coach dashboard
- Exercise library with form guidance
- API access for third-party integration
- Priority support and extended warranty

15.4 Intellectual Property Strategy

15.4.1 Potential Patents

1. **Dual-trigger rep segmentation:** Stillness + extremum detection for boundary identification.
2. **Back-dating algorithm:** Minimum-motion sample search for precise boundary refinement.
3. **Per-rep batch smoothing:** Boundary-anchored drift removal technique.
4. **NPN level-shifter sync circuit:** Cost-effective 1.8V-to-3.3V conversion.

15.4.2 Patent Filing Strategy

- Provisional patents filed before public disclosure
- PCT international application within 12 months
- Target markets: US, EU, China, Japan

15.4.3 Trade Secrets

- Filter tuning parameters (VQF gains)
- Calibration procedures
- Manufacturing test fixtures

15.5 Regulatory Compliance

15.5.1 FCC Certification

ESP32 radio requires FCC Part 15 certification:

- Testing cost: \$5,000-10,000
- Timeline: 8-12 weeks
- Relaxation possible with pre-certified ESP32 module

15.5.2 CE Marking (Europe)

- RED directive (radio equipment)
- RoHS compliance (lead-free)
- Testing cost: EUR 5,000-8,000

15.5.3 FDA Considerations

If marketed for rehabilitation or medical monitoring:

- Class I device (general wellness exemption may apply)
- 510(k) premarket notification if medical claims made
- QSR compliance (quality system regulation)

15.5.4 Data Privacy

- GDPR compliance for EU customer data
- CCPA compliance for California residents
- Minimal data collection policy (anonymized analytics only)

15.6 Manufacturing Strategy

15.6.1 PCB Assembly

Factors sourcing to EMS provider:

- Volume threshold: >1,000 units economically outsourced
- Lead time: 4-6 weeks
- Cost: \$8-12 per board at 10k volume

15.6.2 Enclosure Molding

Table 15.3: Injection molding economics

Item	Cost
Steel mold (2-cavity)	\$25,000-35,000
Per-unit injection cost	\$1.50-2.50
Break-even volume	≈15,000 units

Alternative: 3D-printed enclosures for pilot production (<500 units).

15.6.3 Battery Sourcing

Lithium-polymer cells:

- UL required
- UN38.3 transport compliance
- Cost: \$3-5 per cell at volume

15.6.4 Assembly Test Fixtures

- In-circuit test (ICT) jig: \$15,000-25,000
- Functional test rig (IMU calibration): \$5,000-10,000
- Radio test chamber: \$8,000-15,000

15.7 Cost Structure and Pricing

15.7.1 Bill of Materials (BOM)

Table 15.4: Estimated BOM at 10k volume

Component	Cost
ESP32-WROOM-32	\$2.50
ICM-42688-P	\$3.50
PCB (4-layer, ENIG)	\$2.00
Passive components	\$1.50
Battery (500 mAh)	\$3.00
Enclosure (molded)	\$2.00
Packaging	\$1.50
Assembly/test labor	\$2.00
Total BOM + assembly	\$18.00

15.7.2 Pricing Strategy

Table 15.5: Pricing tiers

Tier	MSRP	Margin
Standard (single sensor)	\$149	≈70%
Pro (2 sensors + app premium)	\$199	≈75%
Team (4 sensors + dashboard)	\$399	≈80%

15.7.3 Distribution Channels

- **Direct-to-consumer (DTC):** Website sales, maximum margin.
- **Gym/wholesale partners:** Volume discounts, installation support.
- **Online marketplaces:** Amazon, CrossFit equipment resellers.

15.8 Software as a Service (SaaS)

15.8.1 Subscription Model

- Free tier: Basic metrics, limited history
- Premium (\$9.99/month): Unlimited storage, advanced analytics
- Team (\$49.99/month): Multi-user dashboard, coach access

15.8.2 Cloud Infrastructure Costs

Per 10,000 active users:

- Compute (AWS EC2): \$500-1,000/month
- Storage (S3 + DynamoDB): \$200-500/month
- Bandwidth: \$100-300/month
- Total: \$800-1,800/month

15.9 Go-to-Market Plan

15.9.1 Phase 1: Beta Launch (Months 1-6)

- 100 beta units to research collaborators and CrossFit boxes
- Feedback collection and firmware iterations
- Case study development

15.9.2 Phase 2: Crowdfunding (Months 6-9)

- Kickstarter campaign with early-bird pricing
- Target: \$50,000 (350 units @ \$149)
- Marketing: performance training influencers, CrossFit media

15.9.3 Phase 3: Public Launch (Months 9-12)

- Official website e-commerce
- Relations with CrossFit affiliate gyms
- Trade show presence (CrossFit Games, FIBO)

15.10 Risk Analysis

15.10.1 Technical Risks

- **Supply chain:** Chip shortages delay production. Mitigation: multi-source critical components, hold safety stock.
- **Battery safety:** Thermal runaway rare but catastrophic. Mitigation: UL-certified cells, protective circuits.
- **Firmware bugs:** Post-release bugs require OTA updates. Mitigation: extensive beta testing, staged rollouts.

15.10.2 Market Risks

- **Competitor price drop:** Established players undercut pricing. Mitigation: cost structure advantage, differentiation via open API.
- **Market slow growth:** VBT adoption slower than projected. Mitigation: diversify into rehabilitation/clinical market.

15.10.3 Financial Risks

Table 15.6: Funding requirements

Expense	Amount
Tooling (molds, test fixtures)	\$75,000
Initial inventory (2,000 units)	\$40,000
FCC/CE certification	\$20,000
Marketing/crowdfunding	\$30,000
Working capital (6 months)	\$150,000
Total seed funding	\$315,000

15.11 Go/No-Go Criteria

Table 15.7: Commercialization decision criteria

Criterion	Threshold
Beta user NPS score	>50
Manufacturing BOM cost	<\$25
FCC certification cost within budget	<\$15,000
Crowdfunding goal met	>\$50,000

15.12 Summary

Commercial translation from research prototype to market-ready product requires careful planning across technical, regulatory, financial, and marketing dimensions. The estimated BOM cost of \$18 establishes a healthy margin at \$149 MSRP. Key success factors include:

- Maintaining accuracy advantage over competitors
- Navigating regulatory certification cost-effectively
- Building initial customer base through beta program
- Total funding need of approximately \$300,000 for initial launch

Chapter 16

Open Source and Reproducibility

16.1 Overview

Reproducibility is a cornerstone of scientific research. This chapter documents the open-source releases, dataset availability, and community engagement strategies that ensure this work can be validated, extended, and built upon by the broader research community.

16.2 Software Releases

16.2.1 GitHub Repository Structure

All software is hosted at `github.com/[username]/vbt_data_collection`:

```
vbt_data_collection/  
  src/                                # C++ host application  
    app/  
    core/  
    gui/  
    processing/  
    sensors/  
  firmware/                           # ESP32 firmware  
    main.cpp  
    imu_driver.cpp  
    espnow_transport.cpp  
  scripts/                             # Python processing pipelines  
    imu_only_pipeline.py  
    orientation.py  
    rep_segmenter_v2.py  
  tools/                               # Annotation studio, loaders  
    annotation_studio/  
  docs/                                # Documentation  
    DEVELOPMENT.md  
    API.md  
    DATASET_GUIDE.md  
  tests/                               # Unit and integration tests
```

CMakeLists.txt
LICENSE
README.md

16.2.2 License

The software is released under the MIT License:

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files, to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software.

This permissive license encourages adoption while requiring attribution.

16.2.3 Versioning

Semantic versioning (SemVer) is used:

- MAJOR.MINOR.PATCH format
- Major: Backward-incompatible changes
- Minor: Backward-compatible features
- Patch: Backward-compatible bug fixes

Current release: v1.0.0 (initial stable release).

16.3 Dataset Availability

16.3.1 Data Hosting

The complete dataset (55 sessions, 622 reps) is hosted on:

- **Figshare:** DOI 10.6084/m9.figshare.xxxxxxx
- **Zenodo:** DOI 10.5281/zenodo.xxxxxxx
- **University repository:** Mirrored for redundancy

16.3.2 Dataset Size

16.3.3 Access Conditions

- **Academic users:** Free access with institutional email verification.
- **Commercial users:** Licensing agreement required.
- **Usage restriction:** Data for research purposes only; no re-identification attempts.

Table 16.1: Dataset storage requirements

Component	Size
Raw IMU CSV (55 sessions)	8.5 GB
Video files (compressed)	45 GB
Depth data (numpy)	12 GB
Annotations (JSON)	25 MB
Metadata (JSON)	5 MB
Total	66 GB

16.4 Documentation

16.4.1 API Documentation

Auto-generated via Doxygen (C++) and Sphinx (Python):

- C++ API: `docs/api/cpp/`
- Python API: `docs/api/python/`
- Host online at: `vbt-data.readthedocs.io`

16.4.2 Tutorials

1. **Quickstart:** 15-minute setup and first session.
2. **Data processing:** Running the IMU-only pipeline.
3. **Annotation guide:** Using the annotation studio.
4. **FPGA deployment:** Building and programming the Artix-7 design.

16.4.3 Code Comments

All public APIs include docstrings. Complex algorithms contain inline comments explaining the mathematical rationale.

16.5 Reproducibility Checklist

To reproduce the validation results in Chapter 9:

1. __ Clone repository: `git clone github.com/.../vbt_data_collection`
2. __ Install dependencies: `pip install -r requirements.txt`
3. __ Download dataset from Figshare.
4. __ Run validation script: `python scripts/validate_pipeline.py -batch`
5. __ Compare output to reference metrics in `tests/reference/`

16.6 Community Engagement

16.6.1 Issue Tracking

GitHub Issues is the primary channel for bug reports and feature requests:

- Respond to issues within 48 hours.
- Triage labels: bug, enhancement, documentation, question.
- Vegan contributors welcome.

16.6.2 Contributing Guidelines

Contributors should:

- Follow the code style (clang-format, black for Python).
- Write tests for new functionality.
- Submit pull requests against the develop branch.
- Include a changelog entry.

16.6.3 Citation Guidelines

When using this work in publications, cite:

Galaxy. (2026). *VBT IMU Data Collection System* [Computer software]. GitHub.
[https://github.com/\[username\]/vbt_data_collection](https://github.com/[username]/vbt_data_collection)

For the algorithm:

Galaxy. (2026). A real-time inertial measurement system for velocity-based training. *PhD Dissertation, University Name*.

16.7 Future Maintenance

16.7.1 Long-Term Ghost Plan

- Archive repository in zenodo.org with versioned snapshots.
- Maintain minimum 1 bug fix release per year for 5 years.
- Deprecation notices with 12-month notice before breaking changes.

16.7.2 Succession Planning

- Designate maintainers in CODEOWNERS file.
- Document build and release procedures.
- Store credentials in secure key management system.

16.8 Summary

Open-source release and dataset availability ensure the research community can:

- Validate reported results.
- Build new algorithms on the same data.
- Extend the system to new use cases.

Transparent documentation and community engagement practices sustain the project beyond the original author.

Chapter 17

Related Work and Comparative Analysis

17.1 Overview

This chapter surveys related academic and commercial work in IMU-based motion tracking, VBT measurement systems, and hardware acceleration of sensor fusion algorithms. The analysis identifies gaps this dissertation addresses.

17.2 Academic Literature

17.2.1 IMU-Based Barbell Tracking

Perez-Castilla et al. (2019)

Found Limits of Agreement ± 0.11 m/s against linear position transducer. Devices were commercial black boxes; algorithm transparency limited.

Gap addressed: Open-source pipeline with reproducible algorithms.

Banyard et al. (2017)

Reported correlation $r = 0.94$ with LPT criterion. However, systematic bias of -0.05 m/s observed at high velocities.

Gap addressed: Sub-0.07 m/s MAE with VQF-based correction.

Laidig et al. (2021)

Established VQF as state-of-the-art for 6-DOF tracking. This work extends VQF to VBT-specific requirements (ZUPT, per-rep smoothing).

Contribution: First application of VQF to weightlifting; validated on 622 reps.

Table 17.1: Complete system comparison

System	Accuracy	Latency	Price	Open
PUSH Band 2.0	0.10 m/s	100 ms	\$299	No
Vmaxpro	0.08 m/s	80 ms	\$399	No
GymAware (LPT)	0.02 m/s	20 ms	\$799	No
Beast Sensor	Unknown	Unknown	\$149	No
This work	0.068 m/s	88 ms	\$150	Yes

17.3 Commercial VBT Devices

17.3.1 Comparative Summary

17.4 Thesis Differentiation

This dissertation contributes:

1. Accuracy comparable to best commercial devices (0.068 m/s MAE)
2. Full open-source release of hardware, firmware, algorithms
3. Comprehensive validation (622 annotated repetitions)
4. Complete FPGA/ASIC migration path
5. Projected \$150 BOM vs. \$300-400 commercial pricing

17.5 Summary

Related work demonstrates growing interest in IMU-based VBT but leaves gaps in transparency, validation rigor, and hardware acceleration. This dissertation addresses these gaps.

Appendix A

System Configuration Files

A.1 ESP32 Firmware Configuration

```
1  # ESP32 Firmware Configuration
2  # File: firmware/config.h
3
4  #define SPI_CLOCK_HZ      10000000    // 10 MHz SPI
5  #define SAMPLE_BATCH_SIZE 8           // Samples per ESP-NOW
6      packet
7  #define ESPNOW_CHANNEL    6           // 2.437 GHz
8  #define ESPNOW_TX_POWER   8           // Max power
9  #define WATCHDOG_TIMEOUT_MS 1000     // 1 second
10 #define IMU_ODR            1000       // 1000 Hz output data
11      rate
12 #define FSYNC_ENABLED      1           // Hardware timestamping
13
14 // IMU Full-Scale Ranges
15 #define ACCEL_FSR          ACCEL_FSR_16G
16 #define GYRO_FSR          GYRO_FSR_2000DPS
17
18 // Calibration Defaults
19 #define CALIB_DURATION_MS  2000       // 2 second static
20      calibration
21 #define CALIB_ACCEL_STDEV  0.02f      // 0.02g threshold
22 #define CALIB_GYRO_STDEV   1.5f       // 1.5 dps threshold
23
24 // VQF Filter Parameters
25 #define VQF_TAU_ACC        2.5f       // Accel weight time
26      constant
27 #define VQF_TAU_BIAS       0.5f       // Bias estimation time
28      constant
29 #define VQF_REST_TH_ACC    0.5f       // Rest detection accel
30      threshold
31 #define VQF_REST_TH_GYR    2.0f       // Rest detection gyro
32      threshold
```

A.2 IMU Register Configuration

```
1  // ICM-42688-P Register Initialization
2  // Reference: Datasheet Section 4.3
```

```

3
4 void imu_init(void) {
5     // 1. Soft Reset
6     imu_write_reg(REG_PWR_MODE, 0x80);
7     delay(50); // Wait for reset to complete
8
9     // 2. Set Clock Source (internal oscillator)
10    imu_write_reg(REG_CLK_CONFIG, 0x01);
11
12    // 3. Configure Gyro
13    // Range: +/-2000 dps, ODR: 1000 Hz, DLPF: enabled
14    imu_write_reg(REG_GYRO_CONFIG0, 0x04); // 4 = 2000 dps
15    imu_write_reg(REG_GYRO_CONFIG1, 0x12); // 1 kHz ODR, 176 Hz
16    // DLPF
17
18    // 4. Configure Accelerometer
19    // Range: +/-16g, ODR: 1000 Hz, DLPF: enabled
20    imu_write_reg(REG_ACCEL_CONFIG0, 0x04); // 4 = 16g
21    imu_write_reg(REG_ACCEL_CONFIG1, 0x12); // 1 kHz ODR, 484 Hz
22    // DLPF
23
24    // 5. Configure FIFO
25    imu_write_reg(REG_FIFO_CONFIG1, 0x08); // FIFO mode enabled
26    imu_write_reg(REG_FIFO_CONFIG2, 0x00); // Sample count = 0 (
27    // unlimited)
28
29    // 6. Configure Interrupts
30    imm_write_reg(REG_INT_CONFIG0, 0x02); // INT1 active high
31    imu_write_reg(REG_INT_CONFIG1, 0x00); // Push-pull output
32
33    // 7. Enable FSYNC timestamping
34    imu_write_reg(REG_FSYNC_CONFIG, 0x20); // FSYNC in FIFO
35
36    // 8. Set Operating Mode
37    imu_write_reg(REG_PWR_MODE, 0x01); // Run mode
38 }

```

A.3 Host Application Configuration

```

1 {
2     "version": "1.0.0",
3     "session": {
4         "data_dir": "datasets/sessions",
5         "max_session_duration_s": 3600,
6         "auto_save_interval_s": 30
7     },
8     "imu": {
9         "sample_rate_hz": 1000,
10        "batch_size": 8,
11        "timeout_ms": 100
12    },
13    "camera": {
14        "serial": "",
15        "depth_fps": 90,
16        "color_fps": 90,

```

```

17     "enable_sync": true
18 },
19 "processing": {
20     "orientation_filter": "vqf",
21     "vqf_params": {
22         "tau_acc": 2.5,
23         "tau_bias": 0.5,
24         "rest_th_acc": 0.5,
25         "rest_th_gyr": 2.0
26     },
27     "zupt_enabled": true,
28     "segmentation": {
29         "still_min_s": 0.08,
30         "peak_window_s": 0.22,
31         "backdate_window_s": 0.3
32     }
33 },
34 "visualization": {
35     "sample_rate_display_hz": 100,
36     "window_duration_s": 10,
37     "show_velocity": true,
38     "show_position": true,
39     "show_reps": true
40 }
41 }

```

Listing A.1: Host Application Configuration JSON

Appendix B

Session Metadata JSON Schema

B.1 Complete Schema Definition (v4.0)

```
1 {
2   "$schema": "http://json-schema.org/draft-07/schema#",
3   "$id": "https://vbt.cs.university.edu/schema/session-v4.json",
4   "title": "VBT Session Metadata",
5   "description": "Schema for velocity-based training session
6     metadata",
7   "version": "4.0.0",
8   "type": "object",
9   "required": [
10     "schema_version",
11     "session_id",
12     "start_time_utc",
13     "exercise",
14     "athlete",
15     "equipment"
16   ],
17   "properties": {
18     "schema_version": {
19       "type": "string",
20       "pattern": "^\\d+\\.\\d+\\.\\d+$",
21       "example": "4.0.0"
22     },
23     "session_id": {
24       "type": "string",
25       "pattern": "^session_\\d{14}$",
26       "description": "Timestamp-based unique identifier"
27     },
28     "start_time_utc": {
29       "type": "string",
30       "format": "date-time",
31       "description": "ISO 8601 formatted timestamp"
32     },
33     "exercise": {
34       "type": "string",
35       "enum": [
36         "back_squat",
37         "bench_press",
38         "deadlift",
39         "snatch",
```

```

39         "clean",
40         "barbell_row",
41         "clean_and_jerk",
42         "front_squat"
43     ]
44 },
45 "athlete": {
46     "type": "object",
47     "required": ["subject_id"],
48     "properties": {
49         "subject_id": {"type": "string"},
50         "sex": {"type": "string", "enum": ["M", "F", "O"]},
51         "age_years": {"type": "integer", "minimum": 18, "maximum":
52             80},
53         "height_cm": {"type": "number", "minimum": 140, "maximum":
54             220},
55         "weight_kg": {"type": "number", "minimum": 40, "maximum":
56             200},
57         "training_experience_years": {"type": "number"}
58     }
59 },
60 "training_context": {
61     "type": "object",
62     "properties": {
63         "mesocycle": {"type": "string"},
64         "week_number": {"type": "integer"},
65         "session_number": {"type": "integer"},
66         "exercise_order": {"type": "integer"},
67         "set_number": {"type": "integer"},
68         "target_reps": {"type": "integer"},
69         "load_kg": {"type": "number", "minimum": 0},
70         "rpe": {"type": "number", "minimum": 1, "maximum": 10},
71         "rest_before_set_s": {"type": "number"}
72     }
73 },
74 "equipment": {
75     "type": "object",
76     "required": ["imu_serial", "camera_serial"],
77     "properties": {
78         "barbell": {"type": "string"},
79         "collars": {"type": "string"},
80         "imu_serial": {"type": "string"},
81         "imu_firmware_version": {"type": "string"},
82         "camera_serial": {"type": "string"},
83         "camera_firmware_version": {"type": "string"}
84     }
85 },
86 "environment": {
87     "type": "object",
88     "properties": {
89         "location": {"type": "string"},
90         "temperature_c": {"type": "number"},
91         "humidity_percent": {"type": "number"}
92     }
93 },
94 "data_files": {
95     "type": "object",
96     "properties": {

```



```

94     "imu_csv": {"type": "string"},
95     "video_mp4": {"type": "string"},
96     "depth_npy": {"type": "string"},
97     "annotations_json": {"type": "string"}
98   }
99 },
100 "quality_flags": {
101   "type": "object",
102   "properties": {
103     "imu_data_complete": {"type": "boolean"},
104     "camera_data_complete": {"type": "boolean"},
105     "sync_verified": {"type": "boolean"},
106     "annotations_verified": {"type": "boolean"}
107   }
108 }
109 }
110 }

```

Listing B.1: Complete Schema Definition JSON

Appendix C

Data File Formats

C.1 IMU Raw Data CSV Format

```
unified_time_s,esp_timestamp_us,host_timestamp_us,  
  accel_x_g,accel_y_g,accel_z_g,gyro_x_dps,gyro_y_dps,gyro_z_dps,fsync_tag  
1715342051.234567,1205234567,1715342051234890,  
  0.012,0.008,1.002,0.5,-0.2,0.1,0  
1715342051.235567,1205235567,1715342051235891,  
  0.015,0.010,1.005,0.6,-0.1,0.2,0
```

Column Definitions:

unified_time_s Wall-clock timestamp aligned to camera frame

esp_timestamp_us ESP32 hardware timestamp

host_timestamp_us Host receive timestamp

accel_x_g ... accel_z_g Accelerometer in g-force units

gyro_x_dps ... gyro_z_dps Gyroscope in degrees/second

fsync_tag 1 if FSYNC edge detected, 0 otherwise

C.2 Rep Segmentation JSON Format

```
1 {  
2   "session_id": "session_20260510_121411",  
3   "exercise": "snatch",  
4   "schema_version": "2.0.0",  
5   "annotations": [  
6     {  
7       "rep_id": 1,  
8       "set_id": 1,  
9       "phases": {  
10        "rest_start": {  
11          "time_s": 1715342051.234,  
12          "position_m": 0.92  
13        },
```

```

14     "concentric_start": {
15         "time_s": 1715342051.434,
16         "velocity_mps": 0.0
17     },
18     "eccentric_start": {
19         "time_s": 1715342052.156,
20         "velocity_mps": 0.0,
21         "peak_velocity_mps": 1.42
22     },
23     "rest_end": {
24         "time_s": 1715342052.890,
25         "position_m": 0.91
26     }
27 },
28 "metrics": {
29     "concentric_duration_s": 0.722,
30     "eccentric_duration_s": 0.734,
31     "peak_concentric_velocity_mps": 1.42,
32     "mean_concentric_velocity_mps": 0.98,
33     "peak_eccentric_velocity_mps": -1.28,
34     "vertical_rom_m": 0.45,
35     "rom_3d_bbox_m": 0.52
36 },
37 "quality": {
38     "detection_method": "stillness",
39     "backdated": true,
40     "confidence": 0.95,
41     "verified_by_human": true
42 }
43 }
44 ],
45 "set_summary": {
46     "set_id": 1,
47     "load_kg": 70,
48     "total_reps": 5,
49     "mean_peak_velocity_mps": 1.38,
50     "mean_concentric_velocity_mps": 0.95,
51     "velocity_loss_percent": 4.2,
52     "time_under_tension_s": 6.8
53 }
54 }

```

Listing C.1: Rep Segmentation JSON Format

Appendix D

Bibliography (Complete)

D.1 Primary References

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